Empower

Intracavity-Doubled, Diode-Pumped Nd:YLF Laser Systems

User's Manual

This laser product complies with performance standards of United States Code of Federal Regulations, Title 21, Chapter 1 - Food and Drug Administration, Department of Health and Human Services, Subchapter J - Parts 1040.10 (a), (1) or (2), as applicable.

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A Division of Newport Corporation

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Preface

This manual provides information regarding the operation and maintenance the *Empower-15* and *Empower-30* laser systems.

The "Introduction" contains a brief description of the *Empower* systems.

The next chapter contains important information on safety. The *Empower* is a Class IV laser and, as such, emits laser radiation which can permanently damage eyes and skin. This section contains information about these hazards and offers suggestions on how to safeguard against them. To minimize the risk of injury or expensive repairs, be sure to read this chapter—then carefully follow these instructions.

"Laser Overview" contains a short section on laser theory regarding the Nd:YLF laser crystal, Q-switching, and second harmonic generation as they are used in the *Empower* laser. This section is followed by a more detailed description of the *Empower* systems and concludes with system specifications.

The next few chapters describe the *Empower* controls, then help you prepare for its installation. Chapter 6, "System Operation," describes how to operate the laser using the *Empower* GUI control software.

The last part of the manual includes a troubleshooting guide and a list of world-wide Spectra-Physics service centers you can call if you need help. *Do not attempt repairs yourself while the unit is still under warranty*; instead, report all problems to Spectra-Physics for warranty repair.

This product has been tested and found to conform to the provisions of Directive 73/23/EEC, the low-voltage directive governing product safety, and the provisions of EMC Directive 89/336/EEC for electromagnetic compatibility. Refer to the "CE Declaration of Conformity" statement in Chapter 2, "Laser Safety," for a complete list of test specifications.

This product conforms to the requirements of 21 CFR 1040.10 CDRH and is compliant to Underwriters Laboratory UL1950 and is listed as ULR for recognized components. This equipment has been designed and tested to comply with the limits for a Class A digital device pursuant to Part 15 of the FCC Rules.

Every effort has been made to ensure that the information in this manual is accurate. All information in this document is subject to change without notice. Spectra-Physics makes no representation or warranty, either express or implied, with respect to this document. In no event will Spectra-Physics be liable for any direct, indirect, special, incidental or consequential damages resulting from any defects in this documentation.

Should you experience any problems with any equipment purchased from Spectra-Physics, or you are in need of technical information or support, please contact Spectra-Physics as described in Chapter 7 “Troubleshooting and Customer Service.”

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If you encounter any difficulty with the content or style of this manual, please let us know. The last page is a form to aid in bringing such problems to our attention.

Thank you for your purchase of Spectra-Physics instruments.

Environmental Specifications

CE Electrical Equipment Requirements

For information regarding the equipment needed to provide the electrical service listed in "Required Utilities" on page 1, please refer to specification EN-309, "Plug, Outlet and Socket Couplers for Industrial Uses," listed in the official *Journal of the European Communities.*

Environmental Specifications

The environmental conditions under which the laser system will function are listed below:

Indoor use

Altitude:	up to 3000 m
Temperatures:	18°C to 35°C
Maximum relative humidity:	85% non-condensing for temperatures in the range above
Mains supply voltage:	do not exceed ± 10% of the nominal voltage
Insulation category:	II
Pollution degree:	2

FCC Regulations

This equipment has been tested and found to comply with the limits for a Class A or B digital device pursuant to Federal Regulations CFR 47 Part 15 subpart A. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

Modifications to the laser system not expressly approved by Spectra-Physics could void your right to operate the equipment.

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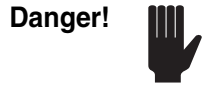
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Warning Conventions

The following warnings are used throughout this manual to draw your attention to situations or procedures that require extra attention. They warn of hazards to your health, damage to equipment, sensitive procedures, and exceptional circumstances. All messages are set apart by a thin line above and below the text as shown here.



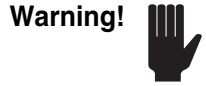
Laser radiation is present.



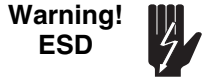
Condition or action may present a hazard to personal safety.



Condition or action may present an electrical hazard to personal safety.



Condition or action may cause damage to equipment.



Action may cause electrostatic discharge and cause damage to equipment.



Condition or action may cause poor performance or error.



Text describes exceptional circumstances or makes a special reference.



Do not touch.



Appropriate laser safety eyewear should be worn during this operation.



Refer to the manual before operating or using this device.

Standard Units

The following units, abbreviations, and prefixes are used in this Spectra-Physics manual:

Quantity	Unit	Abbreviation
mass	kilogram	kg
length	meter	m
time	second	s
frequency	hertz	Hz
force	newton	N
energy	joule	J
power	watt	W
electric current	ampere	A
electric charge	coulomb	C
electric potential	volt	V
resistance	ohm	Ω
inductance	henry	H
magnetic flux	weber	Wb
magnetic flux density	tesla	T
luminous intensity	candela	cd
temperature	celcius	C
pressure	pascal	Pa
capacitance	farad	F
angle	radian	rad

Prefixes								
tera	(10^{12})	T	deci	(10^{-1})	d	nano	(10^{-9})	n
giga	(10^{9})	G	centi	(10^{-2})	c	pico	(10^{-12})	p
mega	(10^{6})	M	mill	(10^{-3})	m	femto	(10^{-15})	f
kilo	(10^{3})	k	micro	(10^{-6})	μ	atto	(10^{-18})	a

Unpacking and Inspection

Unpacking Your Laser

Your *Empower* laser was packed with great care, and its container was inspected prior to shipment. If any major damage was noticed at the time of receipt (holes in the containers, water damage, crushing, etc.), please notify the carrier.

Warning! 

It is recommended that you wait for your Spectra-Physics representative to unpack your system. In no event should you attempt to install the laser yourself, or remove the lid sealing the laser cavity. Either action, if unauthorized, will void your warranty, and you will be charged for the repair of any damage that may result. Refer to Chapter 5, "System Installation."

If you choose to unpack your laser system, carefully inspect the system as you do so. If any damage is evident, such as dents or scratches on the covers or broken parts, immediately notify the carrier and your Spectra-Physics sales representative.

Keep the shipping containers. If you file a damage claim, you may need them to demonstrate that the damage occurred as a result of shipping. If you need to return the system for service at a later date, the specially designed container assures adequate protection.

The *Empower* system is shipped in four crates:

- *Empower* Laser Head
- *Model 2800* Power Supply
- Computer and accessory kit
- Chiller

Each should be carefully opened and unpacked in a clean, dry area. Inspect each component as you unpack it. To open each crate, remove the clamps around the top edge of the crate and lift off the top. (The sides are permanently fixed to the pallet.)

Warning! 

Do not turn the crates upside-down or on their sides. Once the laser and chiller are out of their crates, do not turn them upside-down or on their sides. Doing so can damage these units, and such damage is not covered by your warranty!

It is useful to have a wheeled cart close by to place the laser system on as you remove each component from its create.

The laser head is cradled between foam blocks. Remove any tie-wraps or tape holding the umbilical cables in place, then remove the laser head and its umbilical cables from the crate, taking care not to lift the unit using the umbilical connectors. Place the laser head on the cart. Put the foam blocks back in the shipping crate for storage.

Remove the laptop computer and accessory box and place them on the cart. The *Model 2800* power supply is strapped to its shipping crate. Remove the straps and carefully lift the power supply out of the crate and place it on the cart.

The last crate contains the chiller with the filter and hoses already attached. The chiller is strapped to the base of the crate. Remove the straps and carefully lift the chiller off the base of the crate. DO NOT lift the unit by the filter or its connecting fixtures. Place the unit on the cart.

Wheel the cart to the location where the system is to be installed.

Accessory Kit

- this *Empower User's Manual*
- a laptop computer with control software pre-installed
- one or more beam tube(s)
- serial and power cables for the laptop computer
- a power supply power cable
- Optishield® corrosion inhibitor/algaecide
- a power supply interlock jumper and keys
- spare chiller filter

OptiShield is a registered trademark of Opti Temp, Inc.

Abbreviations

These abbreviations may be found in this manual:

AC	alternating current
AOM	acousto-optic modulator
AR	antireflection
CDRH	Center of Devices and Radiological Health
CW	continuous wave
DC	direct current
E/O	electro-optic
fs	femtosecond or 10^{-15} second
HR	high reflector
IR	infrared
OC	output coupler
ps	picosecond or 10^{-12} second
PZT	piezo-electric transducer
RF	radio frequency
SCFH	standard cubic feet per hour
TEM	transverse electromagnetic mode
Ti:sapphire	Titanium-doped Sapphire
UV	ultraviolet
λ	wavelength

Chapter 1 Introduction

The *Empower* systems are intracavity-doubled, Q-switched, diode-pumped Nd:YLF lasers that produce a pulsed, green, 527 nm output beam. The *Empower-15* can achieve an average power of 15 Watts, while the *Empower-30* is capable of 30 Watts. Output for both lasers is specified at a repetition rate of 5 kHz. At 1 kHz, the *Empower-15* and *Empower-30* can produce pulse energies of 10 mJ or greater and 20 mJ or greater, respectively. The two versions of the system are very similar in design.

Empower output is pulsed at repetition rates between 500 Hz and 10 kHz, as specified by the user and pre-set at the factory. The user also pre-selects whether the laser beam is to exit from the front or side panel of the *Empower* laser head.

The system is controlled by means of custom graphical interface software installed on the laptop computer supplied with the system.

The *Empower* systems consist of four main components:

- the *Empower* laser head
- the *Model 2800* power supply
- a laptop computer with control software installed
- a closed-loop chiller

The Laser Head

The laser head is a sealed aluminum chassis enclosed in an external housing. The housing has an emission indicator and four mounting feet that allow for beam height adjustment. The laser head contains the following components:

- the diode-pumped laser chamber
- the optical resonator assembly
- two acousto-optic Q-switches
- a frequency-doubling crystal and temperature-controlled oven
- a safety shutter
- temperature and laser control electronics



Normal operation will not require that the *Empower* laser head cover to be opened. Removing the laser head cover without prior authorization will void the warranty.

The Model 2800 Power Supply

The *Model 2800* power supply contains the electronics required to drive the diode pump lasers, to Q-switch the laser and to monitor the interlocks. The power supply contains the following components:

- the master control electronics
- the diode laser power supply
- the Q-switch driver
- accessory electronics

The power supply requires a standard, single-phase, 50 or 60 Hz, 110 or 220 Vac 10 A power source. The laser head is connected to the power supply by two removable 3-meter umbilical cables: the Signal umbilical, which provides control and monitoring functions, and the Pump Chamber umbilical, which provides the high levels of current required for the diode pump lasers. In addition, two BNC cables deliver RF power for the Q-switches in the laser head.

The Control Computer

The *Empower* comes with a commercial laptop computer pre-loaded with LabWindows™*-based control software. The system uses an RS-232 serial interface to control and monitor the functions of the laser. Although the particular computer delivered with each laser may vary, it is specified to have:

a high-speed CPU
≥ 256 MB of RAM
≥ 10 GB hard drive
a CD-ROM drive
Windows® 2000** operating system

A copy of the control software is also delivered on floppy disk or CD-ROM.

Caution!

The *Empower* laser is tested with the laptop and control software shipped with the system. Using any other computer or software to control the *Empower* system may cause damage to the laser and may also void the warranty.

The Closed-Loop Chiller

A small closed-loop chiller is provided to dissipate waste heat from the Nd:YLF crystal in laser head, and to stabilize the wavelength of the diode pump lasers to ensure maximum absorption of the pump light in the Nd:YLF crystal. In addition, the chiller circulates cooling water to the frequency-doubling crystal and the two Q-switches. The chiller has two hoses with quick-release connectors and a pressure regulator valve that reduces the water pressure at the laser head. The chiller requires no facility

*LabWindows is a trademark of National Instruments
**Windows is a registered trademark of Microsoft Corporation

water connections. A chemical additive in the distilled water coolant prevents algae growth and corrosion.

All the functions of the system and its components are described in more detail in Chapter 3, “System Overview.”

Chapter 2 Laser Safety

Warning!



This user information is in compliance with Section 1040.10 of the *CDRH Laser Products Performance Standards* from the Health and Safety Act of 1968. Use of controls or adjustments, or performance of procedures other than those specified herein, may result in hazardous radiation exposure.



The *Empower* systems are Class IV—High Power Lasers whose beams are a safety and fire hazard. Take precautions to prevent exposure to direct or reflected beams. Diffuse as well as specular reflections can cause severe eye or skin damage.

This safety section should be reviewed thoroughly prior to operating an *Empower* laser system. Safety precautions listed in this manual should be followed carefully.

General Hazards

Hazards associated with the use of diode-pumped lasers generally fall into the categories listed below. At all times while working with these lasers, please be aware of these potential hazards and act accordingly. You are responsible for your health and of those working around you.

- Exposure to laser radiation can result in damage to the eyes or skin.
- Exposure to chemicals, such as particulate matter or gaseous substances, can be health hazards when they are released as a result of laser material processing or as by-products of the lasing process itself. When these lasers are used to pump dye laser systems, be aware that the dyes used can be extremely hazardous to your health if inhaled or, in some cases, even touched.
- Exposure to high-voltage electrical circuits present in the laser power supply and associated circuits can result in shock or even death.
- Possible health risks are present if pressurized hoses, cylinders, liquids and gasses used in laser systems are damaged or misused.

Precautions For The Safe Operation of Class IV High Power Lasers

- Wear protective eyewear at all times; selection depends on the wavelength and intensity of the radiation, the conditions of use, and the visual function required. Protective eyewear is available from suppliers listed in the *Laser Focus World*, *Lasers and Optronics*, and *Photonics Spectra* buyer's guides. Consult the ANSI and ACGIH standards listed at the end of this section for guidance.
- Maintain a high ambient light level in the laser operation area so the eye's pupil remains constricted, reducing the possibility of damage.
- To avoid unnecessary radiation exposure, keep the protective cover on the laser head at all times.
- Avoid looking at the output beam; even diffuse reflections are hazardous.
- Avoid blocking the output beam or its reflections with any part of the body.
- Establish a controlled access area for laser operation. Limit access to personnel trained in the principles of laser safety.
- Post prominent warning signs near the laser operating area (Figure 2-1).
- Set up experiments so that the laser beam is either above or below eye level.
- Provide enclosures for beam paths whenever possible.
- Set up shields to prevent any unnecessary specular reflections.
- Set up a beam dump to capture the laser beam and prevent accidental exposure (Figure 2-2).

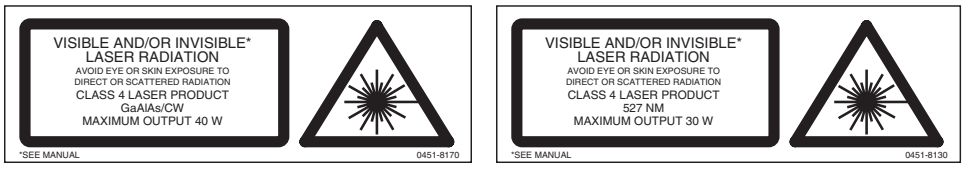


Figure 2-1: These standard safety warning labels are appropriate for use as entry warning signs (EN 60825-1, ANSI Z136.1 Section 4.7).

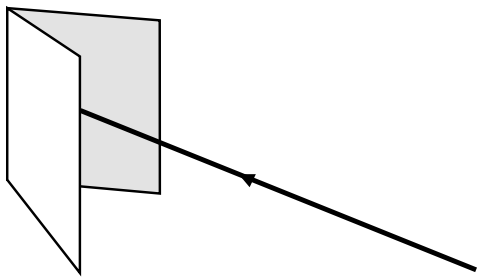


Figure 2-2: Folded Metal Beam Target

Caution!

Use of controls or adjustments, or performance of procedures other than those specified herein may result in hazardous radiation exposure.

Danger!

Operating this laser without due regard for these precautions or in a manner that does not comply with recommended procedures may be dangerous. At all times during installation, maintenance or service of your laser, avoid unnecessary exposure to laser or collateral radiation* that exceeds the accessible emission limits listed in "Performance Standards for Laser Products," *United States Code of Federal Regulations*, 21CFR1040.10(d).

Follow the instructions contained in this manual to ensure proper installation and safe operation of your laser.

Maximum Emission Levels and Protective Eye Wear

It is recommended that laser-safe eye wear be worn at all times when the laser is on. The following are the maximum emission levels possible for this *Empower* system. Use this information for selecting appropriate laser safety eyewear and implementing appropriate safety procedures. These values do not imply actual system power or specifications.

Table 2-1: Maximum Emission Levels

Emission Wavelength	Maximum Power		Pulse
	Empower-15	Empower-30	Duration
527 nm – laser output wavelength	20 W	40 W	100 ns
809 nm – diode laser emission	<< 1 W	<< 1 W	n/a
1053 nm – fundamental wavelength	<< 1 W	<< 1 W	100 ns

During normal operation, the user will not be exposed directly to the diode laser emission or the fundamental wavelength. However, removing the mechanical housing cover will not only invalidate the warranty, but also potentially expose the operator to hazardous levels of laser radiation.

* *Any electronic product radiation, except laser radiation, emitted by a laser product as a result of or necessary for the operation of a laser incorporated into that product.*

Safety Devices

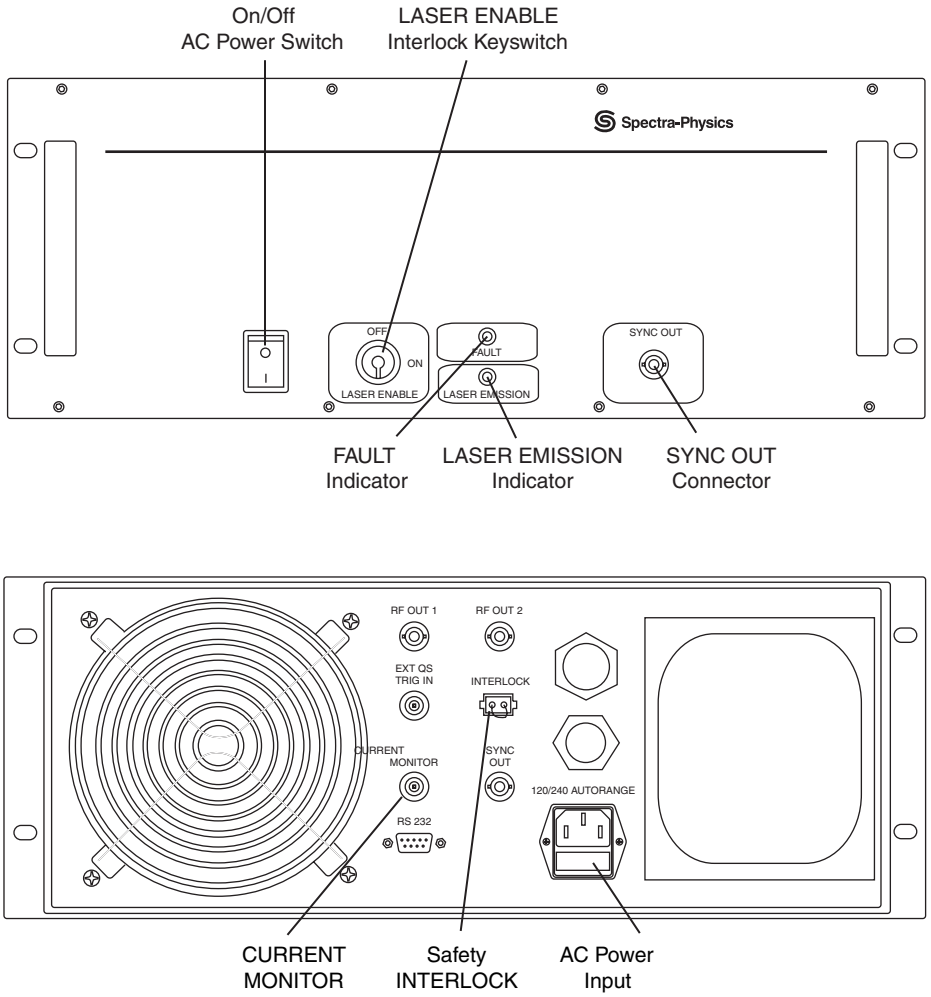


Figure 2-3: Safety Features on the Power Supply

Power Switch

This front panel rocker switch turns on ac power to the power supply. The switch illuminates when in the ON position.

Interlock Keyswitch

The LASER ENABLE keyswitch provides interlock safety to prevent unauthorized personnel from using the system when the key is turned to the OFF position and the key is removed. Turning the key to the ON position closes the interlock and allows the diode laser to be energized if the POWER switch is also on.

The keyswitch also serves as a “reset” switch to clear tripped and corrected interlocks. This key cannot be removed from the switch except in the OFF position. This prevents use of the laser by unauthorized personnel. This keyswitch does not turn off power to the power supply.

Emission Indicators

The green LASER INDICATOR LED on the power supply warns that laser emission is present or imminent. In addition, after a start command has been issued, the white emission indicator on the laser head flashes for approximately 3 seconds before emission is possible to give nearby personnel time to avoid exposure. When this indicator stops flashing (is on steady), diode laser light is present within the pump chamber inside laser head, even though the laser may not be emitting a green laser beam (e.g., when the shutter is closed). All emission indicators remain on as long as the laser is capable of lasing.

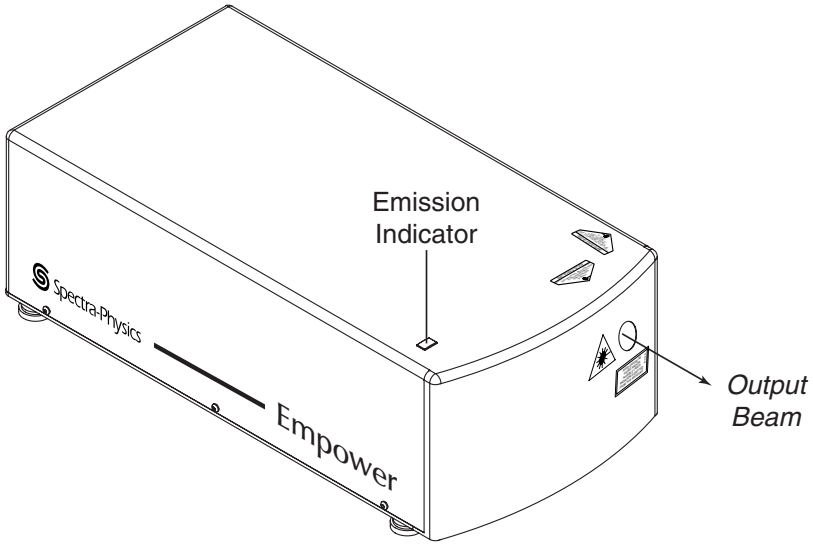


Figure 2-4: *Empower* Laser Head Emission Indicator

Fault Indicator

If the amber FAULT indictor on the power supply illuminates (other than during the start-up sequence as described in Chapter 6), turn off the laser and determine the source of the fault before continuing laser use.

Safety Interlock Connector

The safety INTERLOCK connector on the back of the power supply is used to connect an external CDRH interlock. This interlock, when opened, causes the diode pump lasers to turn off. For example, the interlock connector can be connected to a switch on the door to the laser room that terminates laser action automatically if anyone enters the room. Lasing can resume only after these contacts are once again closed and the keyswitch on the power supply is cycled to clear the interlock function.

A defeating jumper plug is installed at the factory to permit system operation without a safety switch. This plug can be replaced with a similar, nonshorting plug that is wired to a switch designed for a 12 V, 100 mA signal. Wire the external switch "normally closed," so that when the door or safety device opens, the switch opens, turning off the diode lasers and preventing accidental exposure to laser radiation.

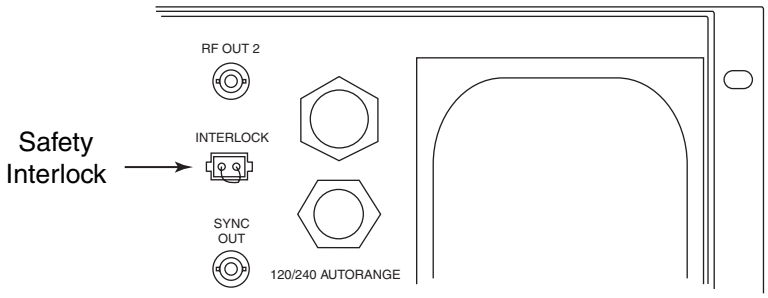


Figure 2-5: External Safety INTERLOCK Connector on the Rear Panel

Safety Shutter

A solenoid-activated safety shutter mounted inside the optical cavity interrupts laser action when necessary. By default, the shutter is closed during start-up (fail-safe mode) and when there is an interlock fault. Only when the laser is turned on by issuing the ON software command is the shutter activated (opened).

Output Ports

The laser beam is set at the factory to exit from either the front panel or the side panel of the laser head per customer specification. The unused port must remain covered at all times.

Protective Housings

Except for a small (<1mW) leakage component present within the green beam, the fundamental 1053nm laser beam is contained within the laser head. As a shield from diode laser light during service, the diode laser pump light is contained within its own separate chamber. This also protects the diode lasers from dust and electrostatic discharge.

Cover Safety Interlocks

Interlock micro-switches ensure that the *Empower* laser head cannot be operated if the interior metal optical cavity cover is not in place.

Warning!

Do not remove the *Empower* interior optical cavity cover. Removing the cover will void your warranty.

CDRH Standards and Computer Control

The *Empower* and its power supply are in comply with CDRH safety standards when operated using the Windows®-based graphical interface supplied with the system. The following features are incorporated into the control software to satisfy CDRH regulations:

- **An emission indicator**—that indicates laser energy is present, imminent or can be accessed. The Main menu of the *Empower* control software displays an obvious black on red "Emergency" indicator when the laser is on. Its presence is required on any control panel that affects laser output.
- **A safety key**—if the power supply interlock key is not accessible, a safety key must be provided to prevent unauthorized use. The password feature of your personal computer (in the CMOS Setup program) or the Windows operating system meets this requirement.

Maintenance in Accordance with CDRH Regulations

This laser product complies with Title 21 of the *United States Code of Federal Regulations*, Chapter 1, subchapter J, parts 1040.10 and 1040.11, as applicable. To maintain compliance with these regulations, once a year, or whenever the product has been subjected to adverse environmental conditions (e.g., fire, flood, mechanical shock, spilled solvent, etc.), check to see that all features of the product identified on the CDRH Radiation Control Drawing (found later in this chapter) function properly. Also, make sure that all warning labels remain firmly attached.

This section lists the maintenance required to keep this laser product in compliance with CDRH regulations.

- Verify all the warning labels listed in Figure 2-6 and Figure 2-7 are present and firmly affixed in the correct locations.
- Verify that removing the INTERLOCK connector on the rear panel of the power supply prevents operation of the laser.
- Verify the time delay between illumination of the emission indicator and start of laser emission gives enough warning to allow personnel to avoid exposure to the radiation.
- Verify the internal beam attenuator (shutter):
  a. operates properly when the laser is turned off from the computer controller,
  b. closes when the keyswitch is turned off, and
  c. blocks access to laser radiation.

Windows is a registered trademark of the Microsoft Corporation.

Radiation Control Drawings

Labels are on page 2-9

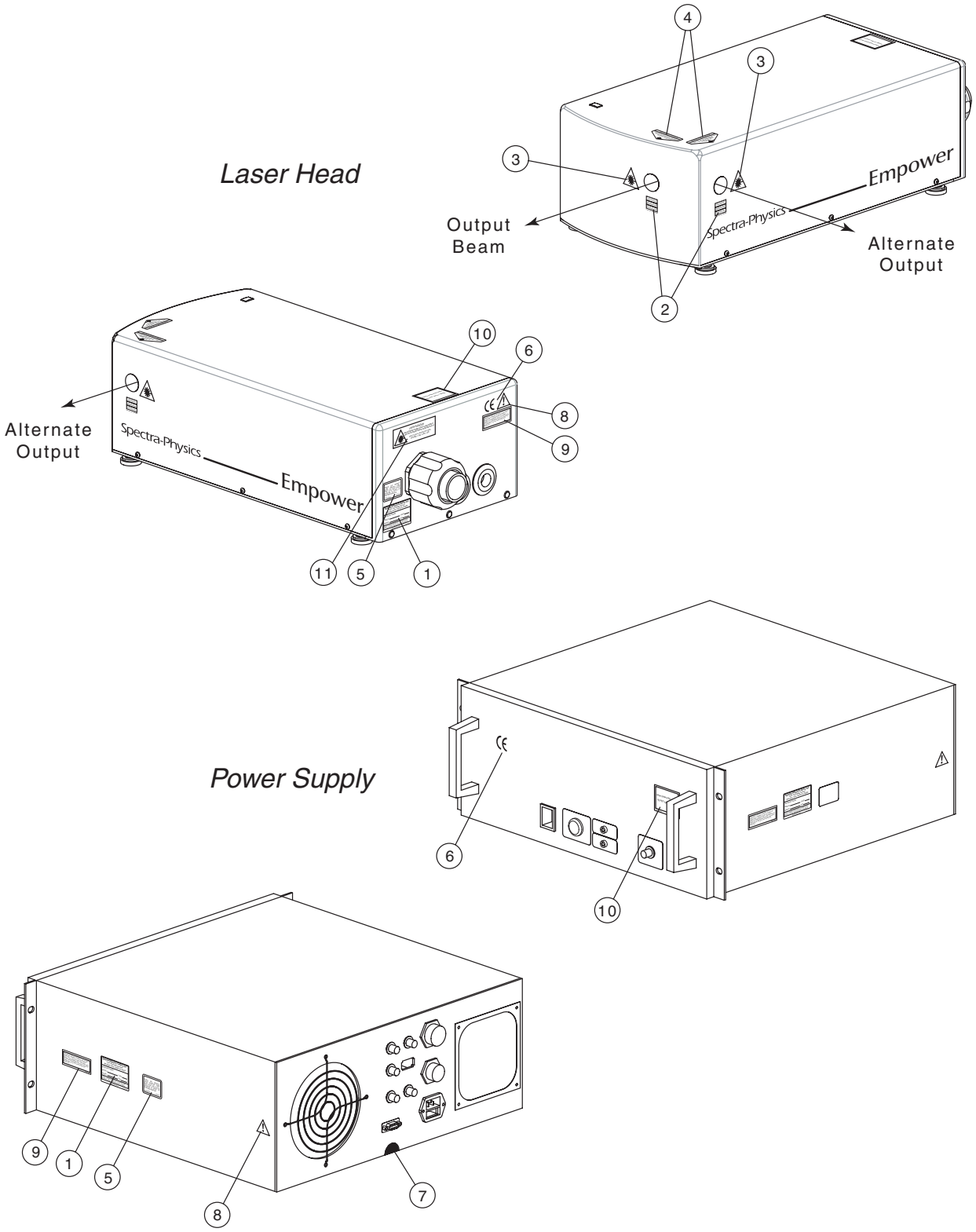
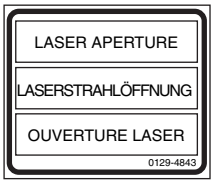


Figure 2-6: *Empower* Radiation Control Drawings

Safety Labels



Model/Serial Number Label
Laser Head/Power Supply (1)



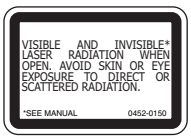
CE Aperture Label (2)



CE Aperture Label (3)



CDRH Aperture Label
Laser Head (4)



Non-interlocked
Housing Label (5)



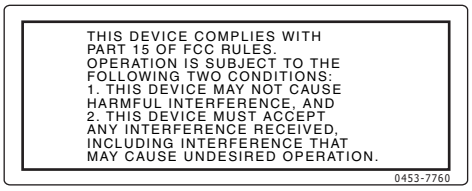
CE Certification
Label (6)



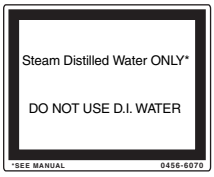
Warranty Seal (7)



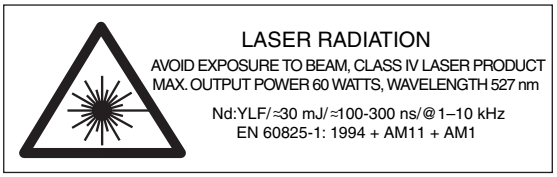
Caution Label (8)



FCC Compliance Label (9)



Distilled Water Use Label (10)



Euro Danger Label (11)

Figure 2-7: *Empower* Safety Labels

Label Translations

For safety, the following translations are provided for non-English speaking personnel. The number in parenthesis in the first column corresponds to the label number listed on the previous page.

Table 2-1: Label Translations

Label #	French	German	Spanish	Dutch
Aperture Label (4)	Un rayonnement visible et/ou invisible est émis par cette ouverture. Ouverture Laser.	Austritt von sichtbarer und unsichtbarer Laserstrahlung! Bestrahlung vermeiden!	Por esta abertura se emite radiación láser visible e invisible; evite la exposición.	Vanuit dit apertuur wordt zichtbare en onzichtbare laserstraling geemitteerd! Vermijd blootstelling!
CE Non-Interlocked Label (5)	Rayonnement laser visible et/ou invisible en cas d'ouverture. Exposition dangereuse de l'oeil et de la peau au rayonnement direct ou diffus.	Beim Öffnen Austritt von sichtbarer und unsichtbarer Laserstrahlung; Bestrahlung von Auge oder Haut durch direkte oder Streustrahlung vermeiden.	Cuando se abre existe Radiación Láser Visible e Invisible; Evite que los ojos y la piel queden expuestos tanto a la radiación directa como a la dispersa.	Zichtbare en niet zichtbare laser-straling wanneer geopend; vermijd blootsteling aan huid of oog aan directe straling of weerkaatsingen.
Distilled Water Label (10)	Eau distillée uniquement. Ne pas utiliser d'eau de-ionisée.	Bitte nur einfach destilliertes Wasser verwenden. Kein de-ionisiertes Wasser!	Solamente use Agua Destilada al Vapor. No use agua sin iónico.	Alleen stoom-gedestilleerd water. Gebruik geen gedeioniseerd water.
Euro Danger Label (12)	Rayonnement laser Exposition Dangereuse. Appareil a laser de Classe IV. Puissance maximum: 60 Watts, Longueur d'onde 527 nm Nd:YLF/≈ 30 mJ/ ≈ 100-300 ns/ @1-10kHz EN60825-1:1994 +AM11+AM1	Laserstrahlung Bestrahlung vermeiden. Laser Klasse IV. Maximale Ausgangsleistung 60 Watts, Wellenlänge 527 nm Nd:YLF/≈ 30 mJ/ ≈ 100-300 ns/ @1-10kHz EN60825-1:1994 +AM11+AM1	Radiación láser Evite la exposición. Producto láser Clase IV. Potencia máxima 60 Watts, Longitud de onda: 527 nm Nd:YLF/≈ 30 mJ/ ≈ 100-300 ns/ @1-10kHz EN60825-1:1994 +AM11+AM1	Laser-straling Vermijd blootstelling! Klasse 4 laser produkt. Max. output vermogen: 60 Watts, Golflengtebereik: 527 nm Nd:YLF/≈ 30 mJ/ ≈ 100-300 ns/ @1-10kHz EN60825-1:1994 +AM11+AM1

CE Declaration of Conformity

We,

Spectra-Physics
1330 Terra Bella Avenue
Mountain View, CA. 94043
United States of America

declare under our sole responsibility that the:

***Empower* diode-pumped laser systems with *Model 2800* power supply,**

manufactured after December 14, 2004,

meets the intent of EMC Directive 89/336/EEC (2004/C 98/05) for electromagnetic compatibility and 73/23/EEC (1973) for low voltage directives. Compliance was demonstrated to the following specifications as listed in the official *Journal of the European Communities*:

EMC Directive 89/336/EEC (2004/C 98/05)

- **EN 61000-6-4: 2001:** Generic Standards—Emission standard for industrial environments
- **EN 55011: 1998+A1 + A2:** Industrial, scientific and medical (ISM) radio-frequency equipment radio disturbance characteristics
- **EN 61000-3-2: 2000:** Limits for harmonic current emissions (equipment input up to and including16 A per phase).
- **EN 61000-3-3: 1995 + A1:** Section 3: Limitation of voltage changes, voltage fluctuation and flicker
- **EN 61000-6-2: 2001:** Generic Standards—Immunity for industrial environments
- **EN 61000-4-2: 1995 + A1 + A2:** Part 4: Section 2: Electrostatic discharge immunity test
- **EN 61000-4-3: 2002 + A1:** Part 4: Section 3: Testing and measurement techniques—radiated, radio frequency, electromagnetic field
- **EN 61000-4-4: 1995 + A1 + A2:** Part 4-4: Testing and measurement techniques—electrical fast transient/burst immunity test
- **EN 61000-4-5: 1995 + A1:** Testing and measurement techniques—surge immunity test
- **EN 61000-4-6: 1996 + A1:** Part 4-6: Testing and measurement techniques—Immunity to conducted disturbances, induced by radio frequency fields
- **EN 61000-4-8: 1993:** Testing and measurement techniques—power, frequency, magnetic field immunity tests
- **EN 61000-4-11: 1994 +A1:** Testing and measurement techniques—voltage dips, short interruptions and voltage variations immunity tests

Low Voltage Directive 73/23/EEC (1973)

- **EN60825-1: Edition 1.2 (2001):** Part 1: Equipment classification, requirements, and users guide, safety of laser products
- **EN60950: (2000 Ed. + Corr 1, 2002):** Safety of Information Technology Equipment
- **EN 61010-1: 2001 + Corr 1 + Corr 2:** Part 1: General requirements—Safety requirements for electrical equipment for measurement

I, the undersigned, hereby declare that the equipment specified above conforms to the above Directives and Standards.



Bruce Craig
Vice President
Spectra-Physics / Newport
December 14, 2004

Sources for Additional Information

The following are some sources for additional information on laser safety standards, safety equipment, and training.

Laser Safety Standards

Safe Use of Lasers (Z136.1:2000)
American National Standards Institute (ANSI)
11 West 42nd Street
New York, NY 10036
Tel: (212) 642-4900

Occupational Safety and Health Administration (Publication 8.1-7)
U. S. Department of Labor
200 Constitution Avenue N. W., Room N3647
Washington, DC 20210
Tel: (202) 693-1999
Internet: www.osha.gov

A Guide for Control of Laser Hazards, 4th Edition, Publication #0165
American Conference of Governmental and
Industrial Hygienists (ACGIH)
1330 Kemper Meadow Drive
Cincinnati, OH 45240
Tel: (513) 742-2020
Internet: www.acgih.org

Laser Institute of America
13501 Ingenuity Drive, Suite 128
Orlando, FL 32826
Tel: (800) 345-2737
Internet: www.laserinstitute.org

Compliance Engineering
Canon Communications, LLC
11444 W. Olympic Boulevard
Los Angeles, CA 90064
Tel: (310) 445-4200

International Electrotechnical Commission
Journal of the European Communities
IEC 60825-1 Safety of Laser Products—Part 1: Equipment classification, requirements and user's guide
Tel: +41 22-919-0211 Fax: +41 22-919-0300
Internet: www.iec.ch

Cenelec
35, Rue de Stassartstraat
B-1050 Brussels, Belgium
Tel: +32 2 519 68 71
Internet: www.cenelec.org

Document Center, Inc.
111 Industrial Road, Suite 9
Belmont, CA 94002
Tel: (650) 591-7600
Internet: www.document-center.com

Equipment and Training

Laser Safety Guide
Laser Institute of America
13501 Ingenuity Drive, Suite 128
Orlando, FL 32826
Tel: (800) 34LASER
Internet: www.laserinstitute.org

Laser Focus World Buyer's Guide
Laser Focus World
Penwell Publishing
98 Spit Rock Road
Nashua, NH 03062
Tel: (603) 891-0123
Internet: lfw.pennet.com/home.cfm

Photonics Spectra Buyer's Guide
Photonics Spectra
Laurin Publications
Berkshire Common
PO Box 4949
Pittsfield, MA 01202-4949
Tel: (413) 499-0514
Internet: www.photonics.com

Chapter 3 System Description

Overview

The heart of the *Empower* systems is a rod of neodymium-doped lithium yttrium fluoride (Nd:YLF, commonly called simply "yilf"). The laser head pump chamber contains the Nd:YLF rod and an array of diode lasers. The diode lasers "pump" the Nd:YLF rod; that is, their output excites the electrons of the neodymium atoms in the YLF crystal, which emit 1053 nm laser light. All the mirrors in the *Empower* laser cavity, including the output coupler, are highly reflective at the 1053 nm wavelength and contain it inside the laser head.

This 1053 nm infrared light is then "frequency-doubled" to 527 nm green light and emitted as energetic Q-switched pulses. Q-switching is a means of concentrating the otherwise continuous optical output of the laser into a series of brief, high-powered pulses that are 100 to 350 ns in duration. The width of these pulses depends on the pulse repetition rate and power level. The repetition rate is set at the factory (as specified by the user at the time of order) at rates between 500 Hz to 10 kHz.

To achieve doubled green output at 527 nm, the Nd:YLF infrared laser light is passed through a lithium triborate crystal, or "LBO." In order to maximize the conversion of the fundamental wavelength to the green harmonic, the LBO crystal is housed in an oven and stabilized at an elevated temperature. The LBO crystal itself offers excellent doubling efficiency and a high threshold for optical damage. The green 527 nm pulses exit the laser head from the forward output port or, if preferred, from the side port.

Nd:YLF Laser Material

The high thermal conductivity of Nd:YLF allows for efficient heat extraction from the gain medium that is essential in high-power, solid-state laser design. In addition, the crystal's natural birefringence overwhelms thermally induced birefringence, eliminating the depolarization problems of optically isotropic hosts like Nd:YAG. Nd:YLF lases at two principle wavelengths: 1047 nm and 1053 nm. The 1053 nm transition is used in the *Empower* because of its much lower tendency to exhibit thermal lensing, another potential problem in high-power systems.

The long upper-state laser lifetime (470 µs) of Nd:YLF provides ample energy storage for Q-switching. The low thermal lensing and natural birefringence of Nd:YLF enable high-power operation without the loss of beam quality or the need for complex resonator designs.

Acousto-Optic Q-Switching

An acousto-optic modulator (AOM) is a block of fused silica that acts as an optical phase grating when vibrated by an ultrasonic wave. The photoelastic effect couples the strain field of the ultrasonic wave to the optical index of refraction in the block. The resultant optical grating has a period and an amplitude set by the acoustic (ultrasonic) wavelength.

When a light beam is incident upon this grating, a portion of its intensity is diffracted out of the beam. By choosing beam parameters properly, any laser beam that attempts to circulate within the resonator experiences a diffraction loss that is sufficient to prevent lasing (i.e., there is no circulating beam). The otherwise low-loss ("high Q") design of the resonator has been switched to a high-loss ("low Q") condition.

With no circulating laser light available to pass through the laser gain medium, the pump energy boosts the gain to a higher level than would otherwise be present. This is where the long lifetime of the upper-state laser level of Nd:YLF is beneficial to the Q-switch process.

The ultrasonic wave is impressed on the AOM by a piezo-electric transducer. Switching off the driving voltage to the transducer returns the AOM to its passive state of high optical transmission, and the laser resonator is returned to its high Q state. The internal beam is no longer deflected, but instead is amplified by the high gain now available in the Nd:YLF rod, and a powerful "Q-switched" laser pulse is emitted.

Voltage is then re-applied to the AOM transducer, which again spoils the cavity Q and allows the gain to rebuild to a high level. This process is repeated at the frequency at which pulsed laser output is desired, taking into consideration other characteristics of the laser.

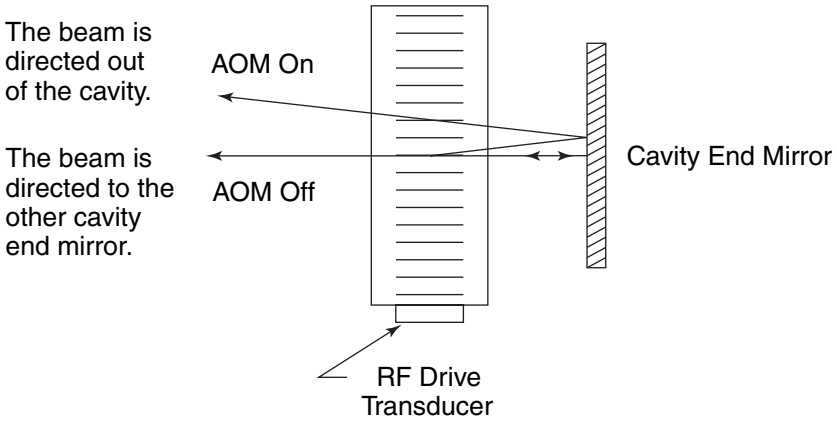


Figure 3-1: An Acousto-Optic Modulator

Q-switching results in energetic pulses ideal for pumping Ti:sapphire ultrafast amplifiers, for materials processing and other similar tasks. The high peak power of Q-switched pulses also greatly enhances the nonlinear optical process of frequency doubling.

Output Power and Beam Quality

The Nd:YLF crystal has lens-like properties that change as the pump energy changes. As a result, the beam size and divergence can vary when laser output power is changed. Moreover, in solid-state lasers, it is common to find that changing pump power effects the pulse width and pulse-to-pulse stability. For example, decreasing pump power can result in a broadening of the pulse width; it may also result in higher pulse-to-pulse instability. Therefore, if output power must be varied without affecting other performance parameters, use an external power control device, such as an acousto-optic modulator. Spectra-Physics can help you select such a device.

Intracavity Frequency Doubling

When an intense laser beam strikes a transparent crystal, a new beam may be produced at twice the frequency of the light of the incident beam. This nonlinear optical effect, referred to both as frequency conversion and as second harmonic generation (SHG), is commonly employed to convert the infrared output from solid-state lasers into a visible, coherent beam.

Since it is the instantaneous power of the fundamental beam that determines the fraction that is converted to the harmonic wavelength, placing the nonlinear crystal *inside* the resonator where it is exposed to high circulating power will increase doubling. Conversion efficiency can also be increased by concentrating the laser energy into pulses using Q-switching. The second harmonic wavelength is coupled out of the resonator by using an output mirror that is 100% reflective at the fundamental wavelength but transmits the second harmonic wavelength.

Frequency doubling requires that the fundamental and second harmonic waves be “phase-matched”; that is, the waves remain in phase with each other over a minimum distance in the crystal. However, since the two wavelengths are substantially different, the fundamental and harmonic beams will experience different values for the index of refraction in the crystal. This wavelength dispersion will cause the two beams to quickly become out of phase unless special techniques are employed that rely on the natural birefringence of the nonlinear crystals.

These crystals possess different refractive indices for different polarization states of the incident light. In frequency doubling, the crystal is oriented so that the fundamental and frequency-doubled beams are orthogonally polarized, with the green beam being the “extraordinary ray.”

Non-critical phase matching relies on the greater temperature dependence of the dispersion of the extraordinary ray. The crystal is heated to a point where the refractive indices for the extraordinary and the ordinary rays at the two wavelengths become equal. Non-critical phase matching is less sensitive to crystal alignment than other phase matching techniques.

Lithium triborate (LBO) is a nonlinear optical crystal characterized by a relatively high damage threshold and a moderate nonlinear coefficient, with excellent material properties. LBO’s small birefringence allows for non-critical phase matching and it provides a larger acceptance angle for highly efficient frequency conversion. The crystal must be heated and temperature-stabilized to maintain good conversion efficiency.

The Empower System

The *Empower* system comprises four basic components:

- the *Empower* laser head
- a *Model 2800* power supply
- *Empower* control software installed on the system computer
- a chiller

The following sections describe the laser head and power supply. The control software is fully described in Chapter 6, "Operation," and the chiller is fully described in its own user's manual.

The Empower Laser Head

Laser Pump Chamber

The laser pump chamber is mounted to the laser head using precision locating pins that allow for ease of replacement. The chamber contains the Nd:YLF laser rod and diode pump lasers. The laser rod is held in a water jacket and flow tube inside the chamber by means of O-rings.

The design of the flow tube and laser rod optimizes the coupling of the energy from the diode lasers into the rod for efficient conversion to infrared laser light. The laser rod and diode lasers are cooled separately by water that enters through manifolds in the bottom of the pump chamber and is split and distributed in parallel paths.

The pump diode laser arrays are arranged in a number of blocks that are mounted around the laser rod in a unique geometry designed to produce uniform pumping throughout the rod volume. Each block of arrays is attached to a single water-cooled heat sink. The water flows through separate parallel channels along the laser rod and over the diode laser heat sinks. The diode lasers are electrically connected, in series, to the diode laser driver via a high-current connector on the pump chamber. This circuit is protected from connection faults by means of a reverse bias diode.

LBO Crystal and Temperature Control

The LBO crystal is anti-reflection (AR) coated for both 1053 nm and 527 nm. It is placed in an oven that maintains a factory-set optimum temperature for non-critical phase matching. A resistive heating element, temperature sensor and heater controller are used to stabilize the crystal temperature. The LBO crystal is constantly stabilized in a narrow temperature range as long as the *Model 2800* power supply ac switch is on. It is recommended that the power supply be left on for this reason.

The heater controller is programmed to limit the rate at which the crystal temperature is changed to prevent the crystal's anti-reflection coatings from cracking. Temperature stability should be achieved within 10 minutes. The temperature controller is a microprocessor-based interlock device. If the LBO temperature is outside the threshold range (set at the factory), an indicator on the software display will turn on and an interlock will prevent laser operation.

AOM Q-Switches

The *Empower-15* uses a single AOM as a Q-switch. In order to hold off its very high circulating intensity, the *Empower-30* cavity uses two synchronized AOMs. The AOM(s) are made from high-quality fused silica to which RF transducers are bonded, and are enclosed in metal housings with coarse azimuthal adjustments. The fused silica is cut and polished to be optically oriented at Brewster's angle for "p" polarized intracavity light.

Approximately 30 Watts of RF power are delivered to the AOM(s) through 50-ohm BNC cables from the *Model 2800* power supply. The AOM(s) are water cooled and have built-in temperature interlocks to shut off RF power if an over-temperature condition occurs. Cooling water flows serially to the AOM(s) and then to the LBO crystal.

The pulse repetition rate of the system is set at the factory to the rate specified by the user at the time of purchase. If an external trigger is used, the rate can be varied by the user ± 20% from the pre-set value, at rates no lower than 500 Hz nor higher than 10 kHz. Larger changes will require a service call.

The Model 2800 Power Supply

The power supply assembly is housed in a rugged steel frame in a 19 in. rack-mount configuration. The power supply contains the master control board, diode power supply, Q-switch RF driver and control electronics.



Potentially lethal voltages and currents are contained in the power supply, and removing the power supply cover will expose these hazards. Contact an authorized service representative before attempting to correct any problem with the power supply.

Master Control PC Board

The master control pc board is a microprocessor-controlled system for setting and monitoring the diode laser power supply, interlocks and Q-switch driver. It is controlled by software through the RS-232 serial interface on the laptop computer that is supplied with the system.

Diode Laser Power Supply

The diode laser power supply, which is controlled and monitored by the master control board, is a self-contained unit that is enclosed in its own metal housing within the power supply. The housing protects both service personnel and the components in the power supply.

Q-Switch Driver

RF power for is provided by a single channel or dual channel driver for the one (*Empower-15*) or two (*Empower-30*) AOMs in the laser head. The driver is a solid-state device consisting of a crystal-controlled oscillator, double-balance mixer, pulse generator with gate and a broadband power amplifier that provides >30 W of drive power for each of the AOM Q-switches. The driver is controlled and monitored by the master control board.

The Chiller and Cooling Water Flow

Spectra-Physics provides closed-loop chillers from a number of different manufacturers that are suitable for use with the *Empower* system. Consult the chiller manual included with the system for specifics about your chiller. Details about the system requirements for cooling water flow are provided in Chapter 5 of this manual.

The water flow in the laser head is initially divided into two parallel paths. One path flows to the pump chamber where it is further parallel-divided to cool the diode pump lasers and Nd:YLF laser rod. The other path cools the AOM(s) first and then the LBO crystal. The heated water flow is combined into one hose and returned to the chiller.

The Control Computer

The *Empower* comes with a commercial laptop computer pre-loaded with Spectra-Physics LabWindows®-based software to control and monitor the functions of the laser. The laptop RS-232 serial interface is used to communicate with the laser system via the power supply. Although the particular computer delivered with each laser may vary, it is specified to have:

a high-speed CPU
≥ 256 MB of RAM
≥ 10 GB hard drive
a CD-ROM drive
a floppy drive
Windows® 2000 or XP operating system

A copy of the control software is also delivered on floppy disk or CD-ROM.

Caution!

The *Empower* laser is tested with the laptop and control software shipped with the system. Using any other computer or software to control the *Empower* system may cause damage to the laser and may also void the warranty.

RS-232 Connection

The control computer is connected to the *Model 2800* power supply using a standard, 9-pin serial cable (provided). By default, the software uses the COM1 printer port on the laptop computer for the built-in serial port.

Control Software

The *Empower* control software is accessed via the Windows Start menu. The appearance of the graphical interface may vary slightly from that shown in this manual, depending on the revision, but all revisions share the same general controls. The latest version of the software is always available from Spectra-Physics.

LabWindows is a registered trademark of National Instruments
Windows is a registered trademark of Microsoft Corporation

Performance Specifications

The table below lists performance specifications for *Empower* laser systems. The environmental specifications are listed in the front of this manual. Utility requirements, including chiller water flow and mechanical specifications are listed in Chapter 5, "System Installation."

Table 3-1: Specifications, *Empower*

Output Characteristics[1]	*Empower-15*	*Empower-30*
Energy per Pulse		
1 kHz	10 mJ	20 mJ
5 kHz	3.0 mJ	6.0 mJ
Beam Diameter	3 mm (nominal)	
Energy Stability[2]	1% RMS	
Polarization	Linear, horizontal	
Transverse Mode[3]	Multimode	
Wavelength	527 nm	
Pre-set Repetition Rate[4]	500 Hz to 10 kHz	

[1] *Due to our continuous product improvement program, specifications may change without notice.*
[2] *RMS with near Gaussian distribution.*
[3] *Optimized for pumping Ti:Sapphire amplifiers.*
[4] *The repetition rate is set at the factory to the value specified by the customer. The repetition rate of an externally triggered system may be varied by ±20% from the factory pre-set value, but at rates no lower than 500 Hz nor higher than 10 kHz.*

Chapter 4 Controls, Indicators, and Connections

Introduction

This section defines the user controls, indicators and connections of the *Empower* laser head and the *Model 2800* power supply. The software used to control the *Empower* system is described in Chapter 6, "Operation." Information on the chiller can be found in the chiller's own user's manual.

The Empower Laser Head

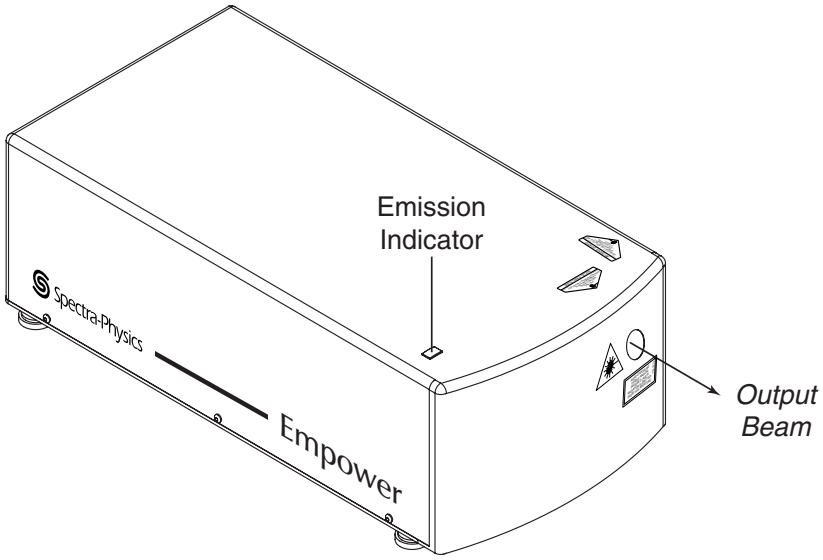


Figure 4-1: The *Empower* Laser Head Emission Indicator

Internal Controls

There are no user-serviceable parts or adjustments inside the *Empower* laser head: no adjustment is required.

Indicators

Emission indicator light—warns of present or imminent laser radiation. This white-light CDRH indicator is located on top of the laser near the output bezel as shown in Figure 4-1. The indicator flashes about 3 seconds before pump light is available to the laser. Following the 3-second delay, the *Empower* will emit laser radiation when the shutter is opened.

Connections

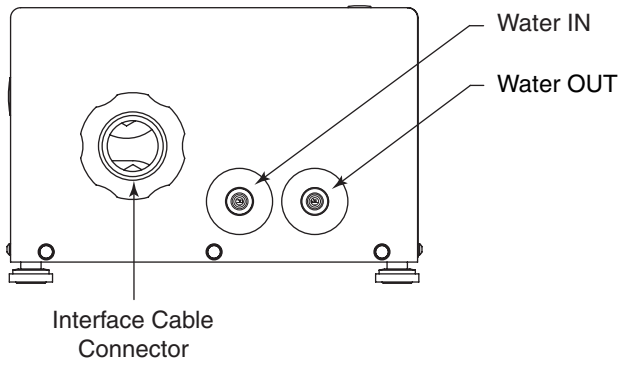


Figure 4-2: Laser Head Connections

Interface Cable Connector—provides permanent attachment for the umbilical from the power supply, which contains the Q-switch cables, the Pump Chamber cable, and the Signal cable. Refer to the descriptions of the power supply connectors for detailed information about these cables. Do not remove the umbilical.

Water IN and Water OUT connectors—provide attachment for the hoses that circulate water to and from the chiller to cool the diode laser pump modules, the Nd:YLF rod, the AOMs and the LBO doubling crystal. To prevent leakage, these connectors automatically close when the hoses are disconnected.

The Model 2800 Power Supply

The power supply contains the analog pc board, the diode power supply, the Q-switch RF drivers and other control electronics.

Power Supply Front Panel

Power switch—enables main ac line voltage to the power supply. The switch illuminates when power is turned on.

LASER ENABLE interlock keyswitch—prevents laser operation when it is set to the off position. Key removal is possible only in the OFF position. This keyswitch is also used to reset latched interlocks by cycling the switch from OFF to ON.

FAULT LED (amber)—flashes when the *Empower* encounters an interlock fault.

LASER EMISSION LED (green)—illuminates when current is available to the diode laser arrays in the *Empower* laser head. Note that the *Empower* is not necessarily emitting laser light when this indicator is illuminated.

SYNC OUT connector (BNC)—provides a TTL output signal that is synchronized with the triggering of the Q-switch.

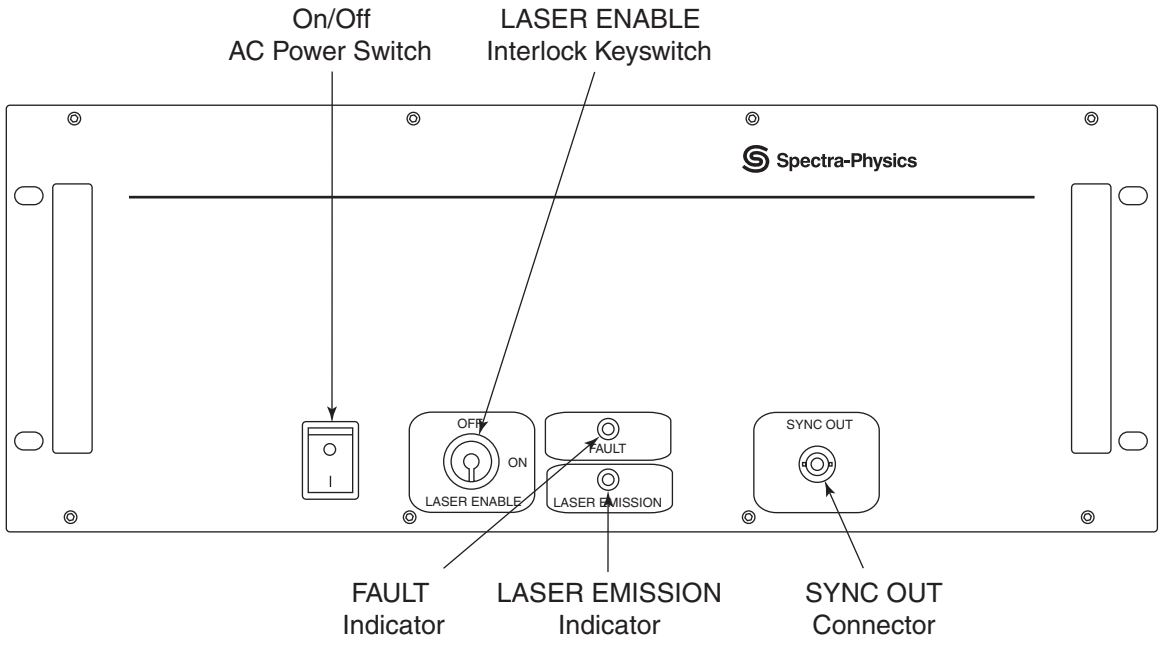


Figure 4-3: The Power Supply Front Panel

Power Supply Rear Panel

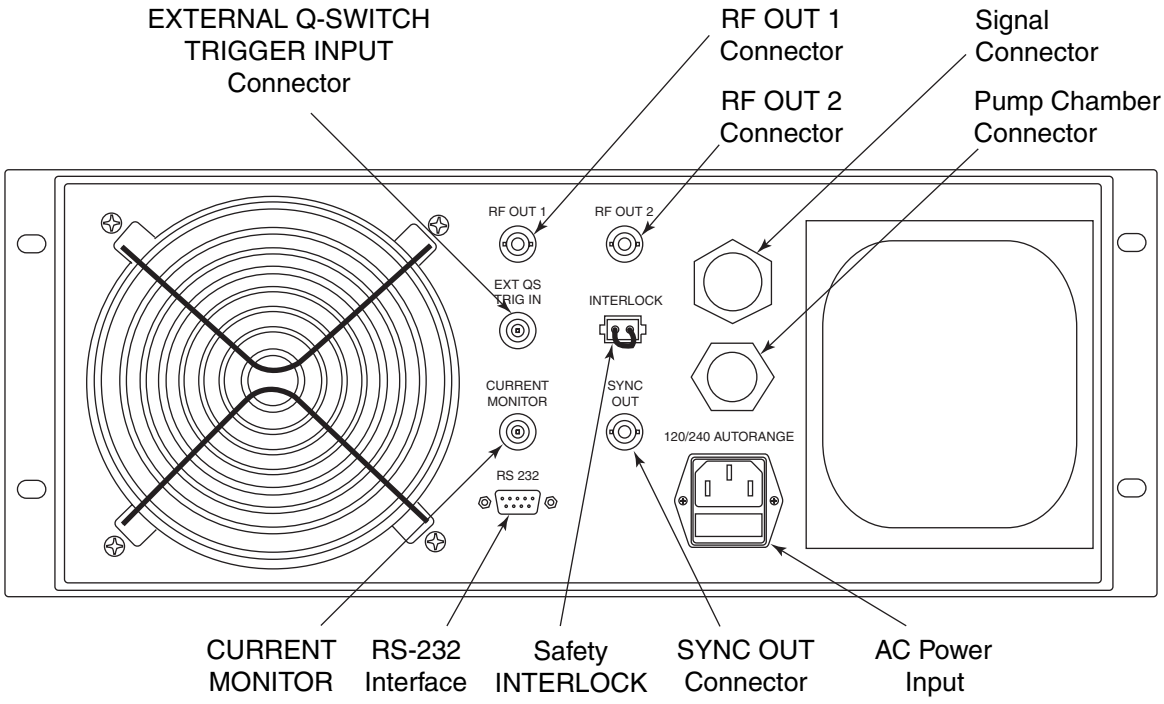


Figure 4-4: The Power Supply Rear Panel

RF OUT connector(s) (BNC)—connect to either one (*Empower-15*) or two (*Empower-30*) 50-ohm BNC cables. Each provides a 30 W nominal (50 W maximum), 50 Vrms sine wave at 27.11 MHz to drive the Q-switches. The AOMs are water cooled and have built-in temperature interlocks to shut off the RF power if an over-temperature condition occurs. DO NOT connect these signals to anything other than the BNC connectors in the *Empower* laser head umbilical.

Danger!



Avoid contact with these terminals if the power supply is on and these terminals are open (not connected to the laser head). Burns caused by RF electrical power are particularly painful.

Signal 26-pin connector (unlabeled)—supplies interlock and control signals to the laser head. Connects to the 26-pin umbilical cable.

Pump chamber 4-pin power connector (unlabeled)—supplies current to the diode laser arrays in the laser head. It connects to the 4-pin connector from the umbilical.

EXT QS TRIG IN connector (BNC)—provides connection to trigger the Q-switch externally within ±20% of the factory preset value from 500 Hz to 10 kHz. Using this mode requires the selection of the "EXT" mode in the control software. If this option is selected, use a TTL signal capable of driving a 50-ohm load.

AC power input—provides connection for the main input power source: either 110 Vac ± 10%, 50/60 Hz, or 220 Vac ± 10%, 50 Hz. The connector includes the fuse holder. Use the fuse size listed in Table 4-1.

Table 4-1: *Empower* Fuse Specifications

Line Voltage/Vac	F1/F2: 3AG Slow Blow
200–240	10 A
100–120	15 A

INTERLOCK connector (2-pin)—is used to provide an optional external interlock for the laser via a controlled access point, such as a laboratory door. If the external interlock is not used, these two contacts must be shorted together before the laser will operate. A defeating jumper plug is installed at the factory to permit operation without a safety switch.

This plug (AMP shell 350777-1, pins 350536-1) can be replaced with a similar, non-shorting plug that is wired to auxiliary safety equipment (such as a door switch) to shut off the laser when actuated (opened). Such a switch must be designed for a 12 V 100 mA signal.

CURRENT MONITOR connector (BNC)—provides an analog output signal to monitor the current flowing through the diode pump laser arrays in the laser head. The scale is 10 Amps/Volt.

RS232 interface connector (9-socket D-sub)—provides connection for RS-232 serial control of the system by the system computer.

Chapter 5 System Installation

The *Empower* system must be installed by a technician trained and authorized by Spectra-Physics to perform these procedures. This section provides information you need to prepare your site for installation. Read the unpacking instructions near the beginning of this manual before unpacking the laser.

Location

Before installation, please select a suitable location for the *Empower* system. Spectra-Physics recommends the laser be located on a stable platform free from vibration. The location must be moderately free from dust and drafts, with low humidity (<50%) and temperature fluctuations less than ±9°F (±5°C) at 77°F (25°C) room temperature.

Review Chapter 2, "Laser Safety," when choosing the location for the laser. Recall that standard practice places the laser so that its beam is considerably below or above eye level, and that specular reflections from surfaces are blocked. Unless your application dictates otherwise, the illumination in the room in which the laser is located should be as bright as practical. It is inadvisable to allow an exposed laser beam to travel a long distance. Please consult with your authorized service representative if necessary.

Required Utilities

Warning! 

The supplied closed-loop chiller is only compatible with the utility service specified at the time the system was ordered. Do not attempt to operate the chiller at a different voltage or frequency under any circumstances.

Table 5-1: Utility Requirements for the *Empower*

Destination	Acceptable Line Voltage/ Frequency (Vac/Hz)	Current Requirements (A)	
		Laser	Chiller
USA	(120 ±12) Vac, 60 Hz	15	20
Japan	(100 ±10) Vac, 50/60 Hz	15	20
	or (200 ±20) Vac, 50 Hz	10	10
Europe	(220 ±22) Vac, 50 Hz	10	10

Preparing to Install the Power Supply

To prepare for the installation of the power supply:

- The power supply must be placed within 10 ft (3 m) of the laser head to avoid straining the umbilicals.
- If possible, the power supply should be placed in a 19-inch rack.
- For proper airflow, allow 6 in. (15 cm) of clearance in front and back of the power supply.
- Do not allow heated air exhausted by the fans on the back panel to return to the air intake, or to flow toward the laser head. Failure to prevent this may cause overheating in the power supply and instability in the laser output.
- Verify the interlock jumper plug is in place on the power supply back panel. If the laser is to be used in a limited access area, remove the jumper inside the plug and wire the terminals to a safety switch.

  Wire the switch so that when the it is activated (e.g., when a door is opened) the switch opens, turning off the laser.
- Verify the local voltage and line frequency are within the acceptable input ranges for the *Empower* and chiller (as ordered).
- Ensure the umbilicals and electrical cables can be safely routed without strain or compression.

Preparing to Install the Laser Head

Warning! 

During installation, it is not necessary to remove the cover that seals the laser head. Doing so will void your warranty, and you will be charged for the repair of any damage that may result.

The laser head is attached to the power supply by a 10 ft (3 m) long umbilical cable. Allow a minimum 3.5 in. (9 cm) of clearance at the back of the laser head for the umbilical bend radius. Two BNC cables from the umbilical connect to the RF OUT connectors on the back of the power supply. The umbilical also connects to the 4-pin connector that supplies current to the Pump Chamber, and to the 26-pin Signal connector that controls and monitors laser output.

Preparing to Install the Control Computer

Place the *Empower* control computer within 10 ft (3 m) of the power supply. The computer should first be set up according to the manufacturer's instructions. Then attach a 9-pin serial cable between the computer's serial port and the RS-232 port on the *Model 2800* power supply.

The *Empower* control software is pre-installed on the control computer.

Mechanical Specifications

Table 5-2: Component Weights

Component	Weight
Laser Head (OEM)	33 lb (15 kg)
Power Supply	45 lb (21 kg)
Chiller (dry, typical)	122 lb (55 kg)
Umbilical Length	10 ft (3 m)

Preparing to Install the Chiller

Spectra-Physics uses chillers provided by a number of different manufacturers that are suitable for use with the *Empower* system. For specific information about the chiller shipped with your system, consult the chiller manual included with the system.

- Verify the local voltage and line frequency are within the acceptable input ranges for the chiller.
- Set the chiller on the floor close enough to the *Empower* that the cooling hoses will reach without strain.

Caution!

The chiller must be located so that the warm air exhausted from the back panel is not drawn into the *Model 2800* power supply. Allowing it to do so could affect the stability of the laser.

- Allow 6 in. (15 cm) of clearance between the chiller and the *Model 2800* power supply to prevent hot air from one unit from being drawn into the other.
- The chiller should not be placed above the laser or power supply. If a leak or condensation ever developed in the chiller, dripping water could damage the unit below it.
- The hose connectors are attached so that the laser head input hose (which is marked with an arrow pointing toward the laser head) is attached to the innermost hose connector on the laser head.

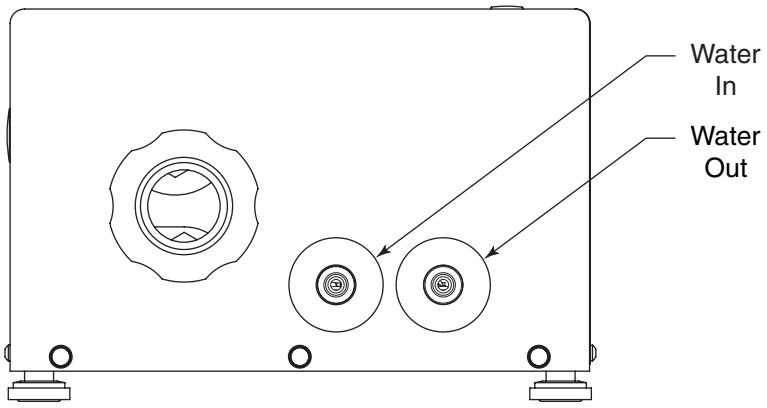


Figure 5-1: Laser Head Water Connections

Caution!

The direction of cooling water flow is important for proper operation of the *Empower* system. The water connectors on the laser head are identical: take care to connect hoses in the proper orientation. Incorrect flow direction will cause an internal flow interlock to turn off the laser.

Danger!

Before handling the OptiShield water treatment chemical, read Appendix A, "OptiShield Additive," which describes the potential hazards of this substance and provides instructions for its use. This chemical can be harmful if swallowed, inhaled or absorbed through the skin or eyes.

Caution!

Use only **STEAM-DISTILLED** water in the *Empower* cooling system. The use of highly de-ionized water may cause corrosion damage.

While the chiller model may vary, system requirements for the chiller are the same for all chiller models. These include:

- a flow rate of at least 2.5 gal (9.5 l) per minute
- water pressure between 50 and 55 psi (345 to 380 kPa)
- water temperature between 65–86°F (18–30°C)

Verify your chiller is operating within these ranges. Refer to the chiller manual included with your *Empower* system if you need to make adjustments to your chiller.

Outline Drawings

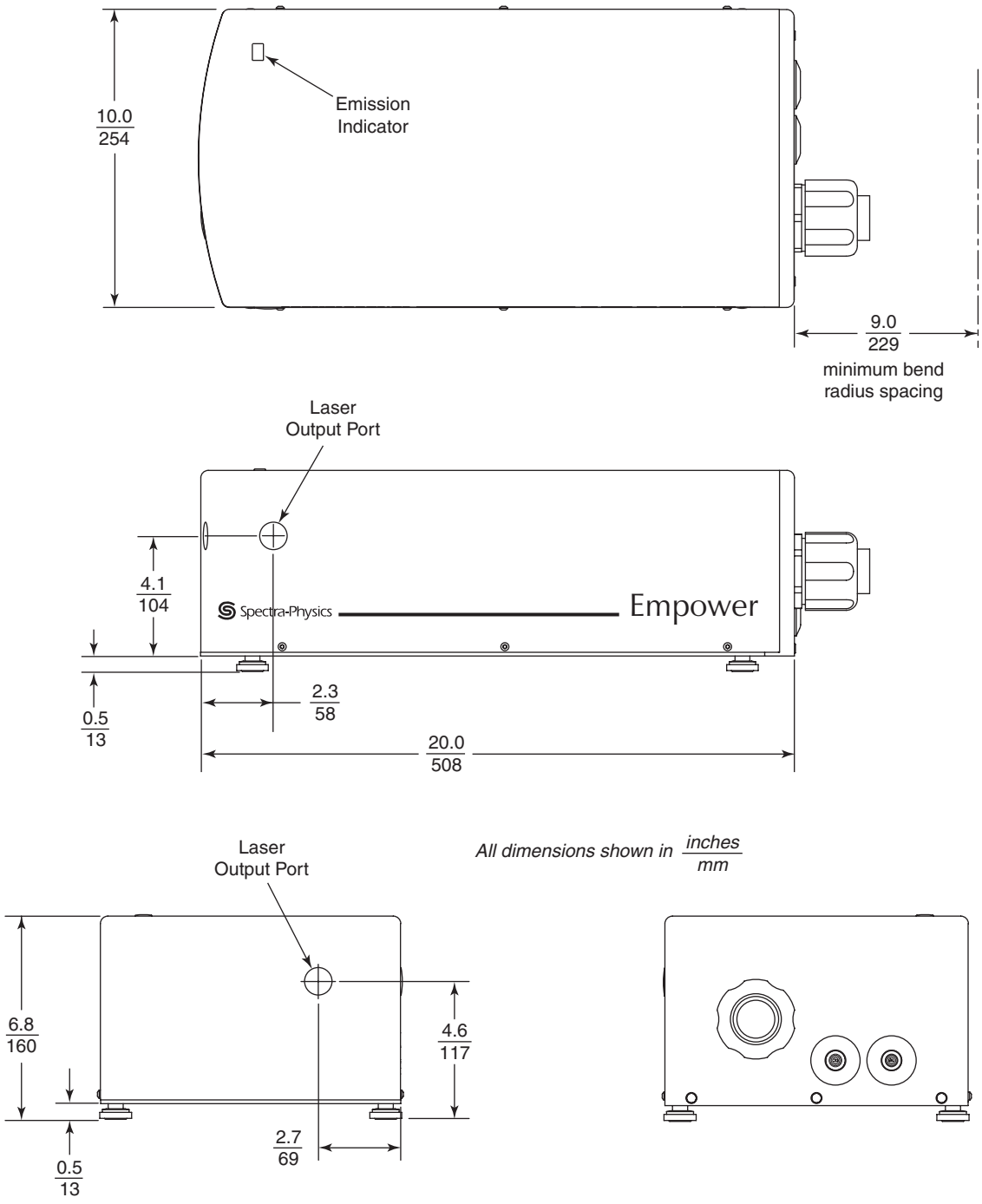


Figure 5-2: *Empower* Laser Head Outline Drawing

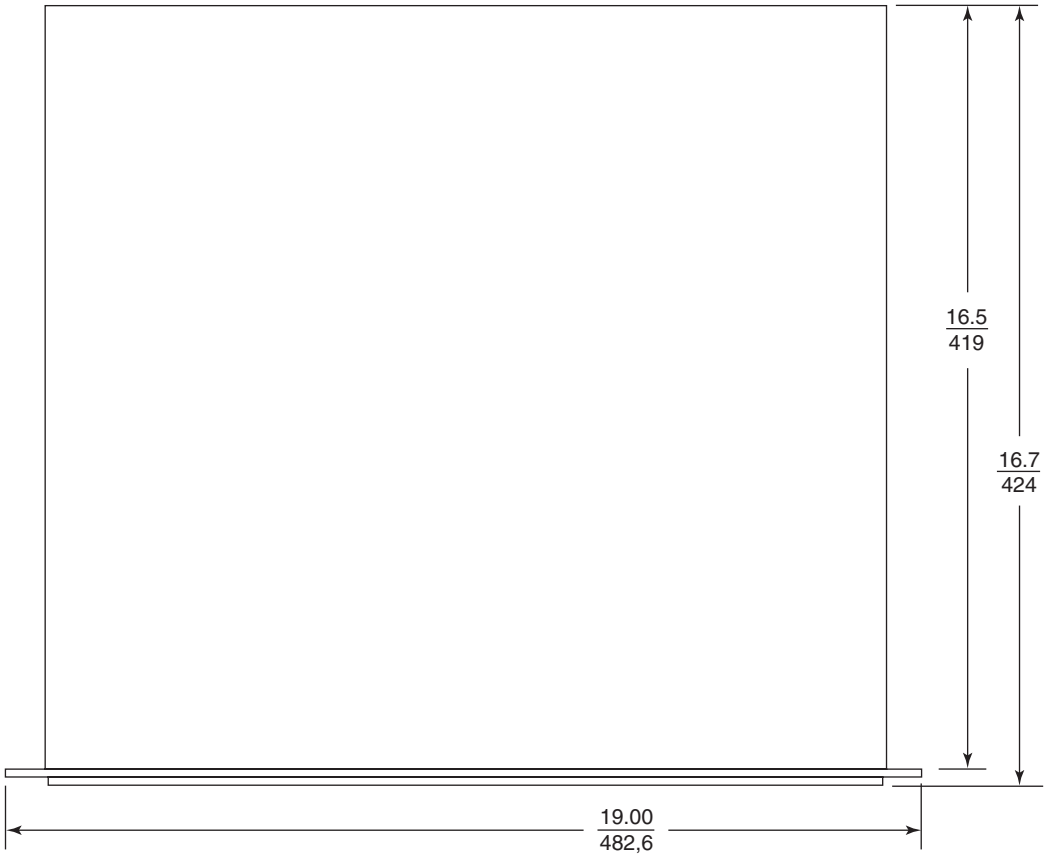


Figure 5-3: *Model 2800* Power Supply Outline Drawing

Chapter 6 System Operation



Safety glasses for all lasing wavelengths must be worn at all times when operating this laser system. For your safety and the safety of others, please read "Chapter 2, Laser Safety." Consult your laser safety officer to select appropriate safety glasses.

Ensure that all persons in the room are wearing adequate eye protection. Also ensure that the anticipated beam is safely terminated into a high power beam block or power meter.

The *Empower* laser system is typically operated using the Spectra-Physics control software pre-installed on the system computer. This software controls both the laser head and the power supply.

If this is your first time operating the *Empower* system, before starting the laser, familiarize yourself with the operation of the power supply and the control software as described in this chapter.

Daily Start-Up Procedure

To turn on the system:

1. Turn on the chiller if it is not already on.
2. Turn on the power supply ac power switch if it is not already on. Allow 15 minutes for the harmonic crystal to reach operating temperature.
3. Boot the control computer and log in (if necessary).

   To prevent unauthorized access to the laser system, the system owner should implement a password requirement on the control computer using either the Windows password feature or the one provided by most computer manufacturers via the BIOS Setup menu.
4. Start the *Empower* control software.

   After the control software begins communicating with the *Empower* and all interlocks are closed, the STOP indicator on the Main menu (Figure 6-1) will turn on, indicating that the laser is ready to be started.

   If the FAULT indicator on the power supply flashes, or the FAULT indicator on the Main menu is on, refer to the Troubleshooting Guide in Chapter 7 for help in clearing the interlock faults.
5. Insert the key in the power supply keyswitch and turn it to the ON position. The FAULT indicator will come on if the harmonic crystals are not at operating temperature. When this happens, turn the keyswitch off and wait an additional 5 minutes, then try again.

6. From the View pull-down menu, select the Setup menu and use the Set Current control to set the diode laser current to a value you would consider a "low" setting (refer to "Diode Laser Current Settings" on page 6-3 for further explanation).
7. Verify the laser beam will be directed into an appropriate termination (a power meter, a beam block, etc.).
8. On the Main menu, click and hold down the green ON button for 5 seconds to start the laser ramping up to the set current. The COUNT LED on the menu will turn on, as will the EMISSION indicator. The white emission indicator on the laser head will begin to flash.
9. After about 3 seconds, the *Empower* laser head will begin producing laser radiation, the COUNT LED on the display will turn off and the emission indicators will glow steadily.
10. It takes the chiller about 15 minutes to stabilize the temperature of the laser head and, thus, the output of the laser. If the laser is used frequently, leaving the chiller on between periods of use will eliminate or greatly reduce this stabilization time.
11. Measure output power with a power meter capable of measuring 30 Watts minimum. Within several minutes, the 527 nm power should be the same as or close to the power that was previously measured at the Low current setting.

Changing Repetition Rates

Caution!

Your *Empower* was optimized at the factory at your requested Q-switch repetition rate. Operating the laser at rates that exceed ±20% of the factory pre-set value or lower than 500 Hz or higher than 10 kHz will result in decreased performance and possible damage to the instrument.

Externally Triggered

If the *Empower* Q-switch is to be externally triggered, a limited range of variation in the repetition rate is available as noted in the caution note above.

To externally trigger the Q-switch, provide a TTL-level signal to the EXT QS TRIG IN connector on the back of the power supply. The signal pulse must be at least 2.5 µs in duration and set to a proper frequency as noted earlier. Q-switch operation is synchronized to the rising edge of the input pulse.

Use the Setup Display menu to change Q-switch parameters. Set the QSW SOURCE setting to EXTERNAL.

Internally Triggered

If the *Empower* Q-switch is internally triggered, set the QSW SOURCE setting to INTERNAL. The repetition rate for the laser was set at the factory to the value specified at the time the system was ordered. Changing the repetition rate of an internally triggered system requires a service call from Spectra-Physics.

System Shutdown

To shut off the system:

1. Click on the STOP button on the Main menu. The shutter will close and the diode lasers will shut off immediately, terminating laser emission.
2. Turn the power supply keyswitch to OFF and remove the key from the keyswitch.
3. Exit the *Empower* control software.
4. Keep the ac power switch on the power supply set to ON.
5. Unless the *Empower* system is used infrequently, allow the chiller to remain on to minimize laser warm-up time the next time it is turned on. If the laser is not to be used for some time, turn off the chiller. The recommended duty cycle for the chiller is to start it in the morning and allow it to run all day, shutting it down at night only if the *Empower* is to be turned off for a long period of time.

Diode Laser Current Settings

The *Empower* laser comes with a set of performance data that lists the factory-measured output power as a function of diode laser current at the Q-switch repetition rate specified when the laser was ordered. Your authorized service representative will verify these measurements when the laser is installed.

To simplify operating and trouble-shooting procedures, it is useful to establish three nominal diode laser current settings for in-house operation: Low, Medium and High. For example, settings of 15, 20 and 25 A typically produce 6, 15, and 20 W output from an *Empower-30* system (at 1 kHz). A Low setting should produce the minimum amount of light required to align the laser to a target, the High setting should produce the highest amount of light typically used to do the job, and the Medium setting can be any arbitrary point in between. The Setup menu under the View pull-down menu provides a control for setting the laser diode drive current. Note that only one setting can be implemented at a time.

The maximum possible current for the diode pump lasers is set at the factory to ensure that the *Empower* will meet its specified power for thousands of hours of operation. After several hundred hours of operation the diode laser output power will decrease slightly. This is normal following a “burn-in” period.

Control Software

The *Empower* system is normally operated using the Spectra-Physics control software pre-installed on the laptop computer provided with the system. The software is accessed via a Desktop icon or from the Programs folder on the Windows® START menu. This section describes software revision 1.1, which was the latest version available when this manual was released. The latest version of the software is always available from Spectra-Physics.

Windows is a registered trademark of the Microsoft Corporation.

The Main Menu

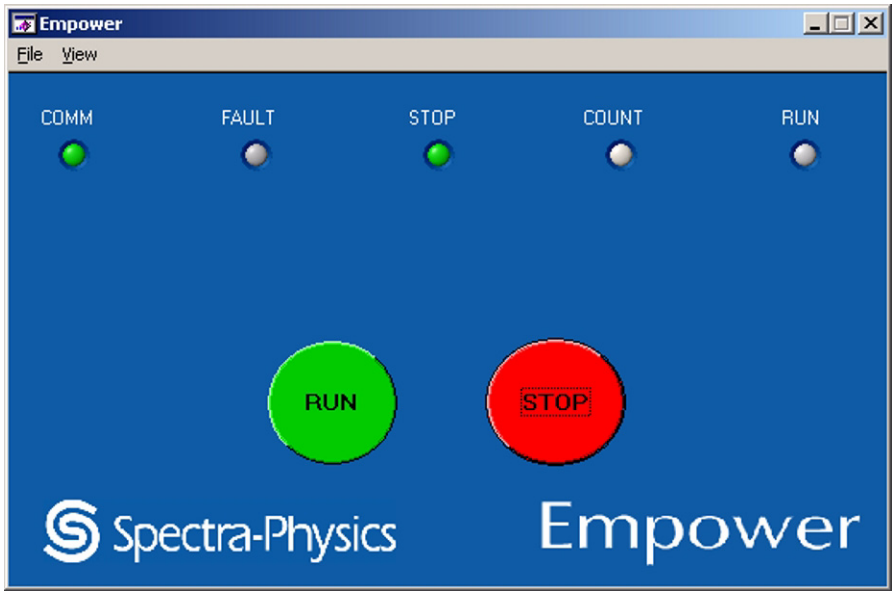


Figure 6-1: The *Empower* Main menu.

RUN and STOP buttons—turn the *Empower* laser on and off.

When the RUN is selected, the diode laser arrays turn on, the RUN indicator turns on and the white emission indicator on the laser head flashes. The COUNT indicator then turns on and the diode laser arrays begin a controlled ramp-up in current (which allows time for the doubling crystal to reach operating temperature). After a delay of 3 seconds, during which time a tone sounds, the *Empower* begins to lase.

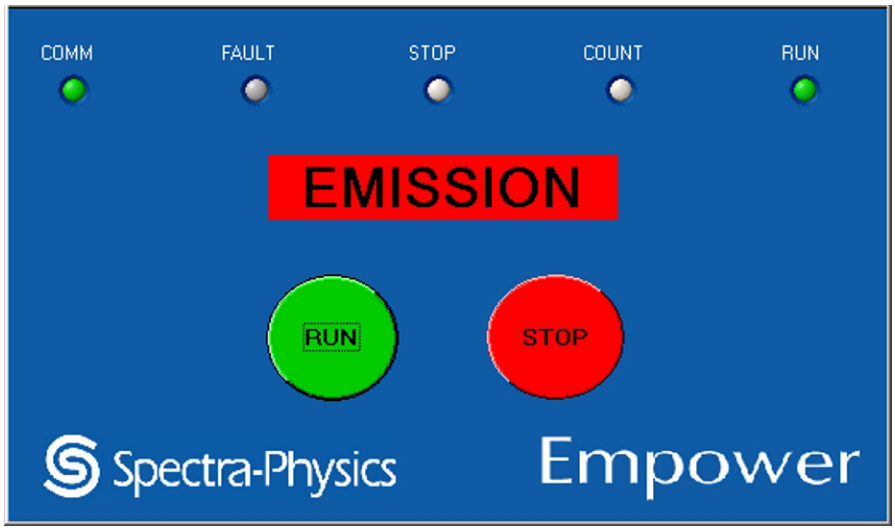


Figure 6-2: The Main menu when the laser is on.

Once lasing occurs, the COUNT indicator turns off and the LASER EMISSION LED on the power supply and the RUN indicator on the Main menu turn on. The emission indicator on the laser head will now glow steadily.

The Main menu also displays “Emission” in the center text bar as shown in Figure 6-2.

When the STOP button is pressed, the *Empower* stops lasing (the pump laser diodes are turned off and the laser cavity shutter closes).

RUN, STOP, and COUNT indicators—are described in the previous section.

FAULT indicator—turns on if a fault is detected. See the description of the Active Faults Display below for more information.

COMM indicator—turns on when the serial link is operating properly. If serial communication is lost, the indicator will turn off and the system will automatically attempt to re-set the link. If unsuccessful, save the data and settings, then exit the *Empower* control software, check the cable connections and restart the software.

The Main menu should remain visible at all times: click on it (or close the other windows) to return to it.

The Information Menu

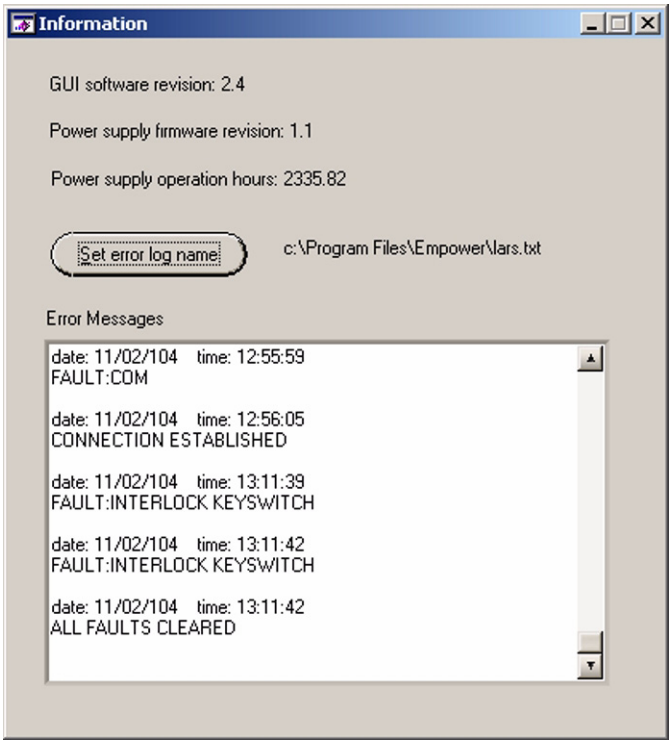


Figure 6-3: The Information Menu

The Information menu is found under the View pull-down menu at the top of the Main menu. This menu displays basic *Empower* status information.

The top portion of the menu lists the revision level of the *Empower* software, along with the revision of the firmware that controls the power supply. Also listed is the accumulated total hours that the power supply has been operated, an important measure of diode pump laser lifetime performance.

SET ERROR LOG NAME button—allows you to name and assign a location for the text log file for fault reporting. The name of the present log file is displayed to the right of the control button.

The log file accumulates fault reports for the entire history of laser operation. The most recent entries appear at the end of the text file. You may edit or clear the log file as you would any text file. The lower text window in the display shows the most recent fault conditions and their time stamps.

The Setup Menu

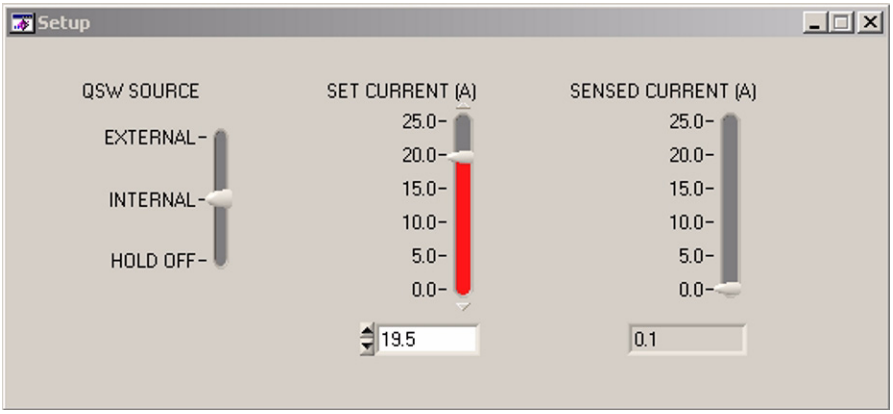


Figure 6-4: The Setup Menu

The Setup menu is found under the View pull-down menu at the top of the Main menu. This menu allows the operator to set the Q-switch trigger source or turn off triggering altogether, to set the laser diode operating current and to see the actual (sensed) operating current.

QSW SOURCE slide control—sets the Q-switch trigger source:

EXTERNAL—selects the EXT QS TRIG IN BNC connector on the back of the power supply as the Q-switch trigger source. The basic repetition rate is set at the factory to the value specified when the system was ordered. However, by selecting EXTERNAL, the system can be varied ±20% from the preset value, for example from 800 Hz to 1.2 kHz for a 1 kHz system. Repetition rates cannot be set at less than 500 Hz nor more than 10 kHz. If set less than 500 Hz, a fault will occur and the laser will turn off. RF power for the Q-switches is provided by the system.

INTERNAL—selects the internal clock as the Q-switch trigger source. The clock is set at the factory for the repetition rate value specified when the system was ordered. RF power for the Q-switches is provided by the system.

Changing the INTERNAL repetition rate requires a Spectra-Physics service call.

HOLD OFF—is used only during service procedures.

Note

Changing the QSW SOURCE setting while the laser is running will turn off the laser. To restart the laser, click on the RUN button again.

SET CURRENT [A] control—sets the current (in amps) for the diode pump lasers up to the factory maximum set point. The numeric value can be entered directly in the lower window or in 0.1 A steps using the slide control.

SENSED CURRENT [A] display—shows actual diode laser current.

The Active Faults Menu

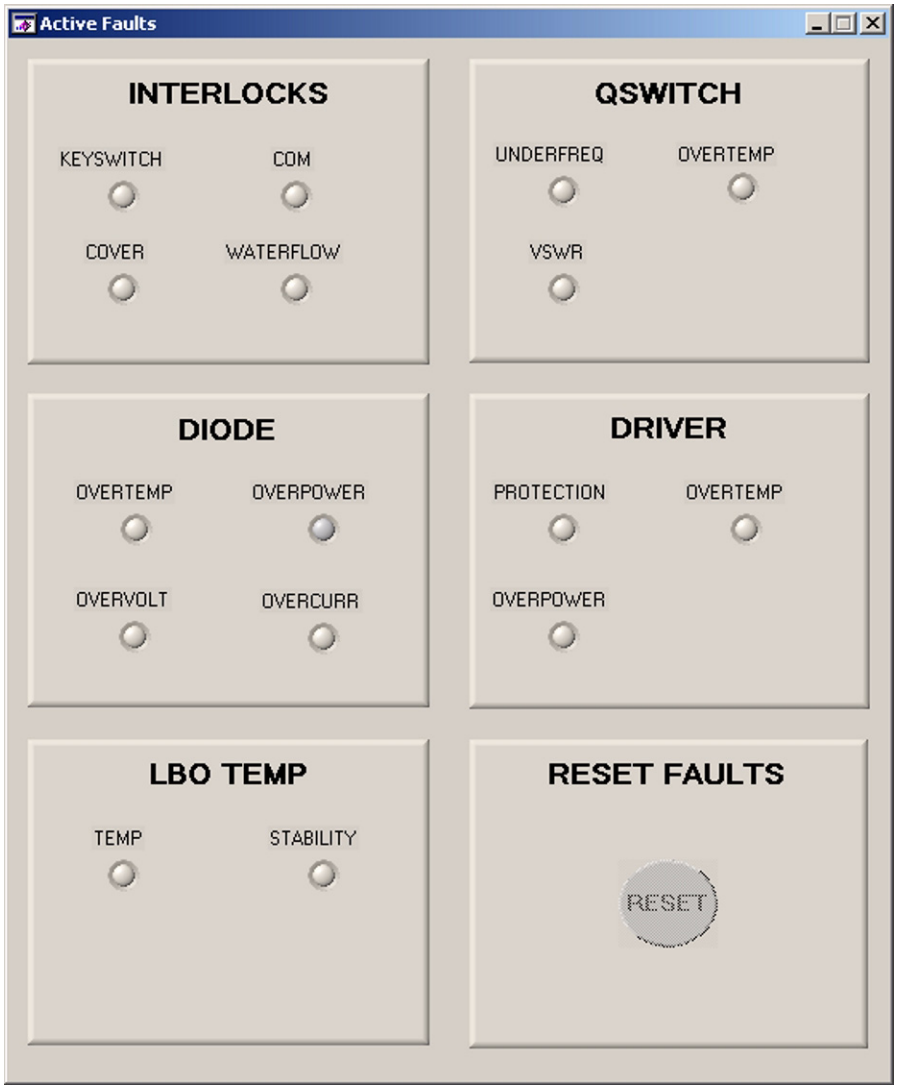


Figure 6-5: The Active Faults Menu

The Active Faults menu is found under the View pull-down menu at the top of the Main menu. This menu represents a total of 16 possible *Empower* system faults displayed as LED-type indicators arranged in 5 categories.

An “LED” is illuminated if the corresponding fault is presently active. All active faults are illuminated.

Use the **RESET** button at the bottom of the menu to clear the fault interlocks once the underlying conditions have been detected and corrected.

The **INTERLOCKS** LEDS turn on when one of the following interlocks is open:

KEYSWITCH—turns on if the keyswitch is in the OFF position.

COM—turns on when a communication interruption is detected. Exit the *Empower* control software, check the serial cable connections and verify that the power supply is turned on, then cycle the ac power on the power supply and restart the *s*oftware to clear the fault.

COVER—turns on when the laser head cover is opened.

Warning!

DO NOT OPEN the laser head! Doing so can seriously compromise the cleanliness of the laser cavity. If the cover has been opened, close it immediately and contact your Spectra-Physics service representative.

This indicator also turns on when the user interlock opens (a more likely scenario). Verify the interlock switch is closed or the interlock jumper is in place, then cycle the keyswitch to clear the fault.

WATERFLOW—turns on whenever chiller water flow is less than 2.0 GPM. Verify the chiller is functioning properly, that the water pressure and reservoir level are correct and that the water hoses are not kinked, then cycle the keyswitch to clear the fault.

The **EMPOWER QSWITCH** LEDS turn on when one of the following Q-switch fault conditions is detected:

UNDERFREQ—turns on when the Q-switch mode is set to EXTERNAL and the external trigger rate drops below 500 Hz. Reset the external Q-switch trigger to produce TTL pulses at 500 Hz or higher. Note: the *Empower* repetition rate can only be varied ±20% from the factory preset value.

OVERTEMP—turns on when either AOM exceeds the safe operating temperature for that device. Verify the chiller is functioning properly, that the water pressure and reservoir level are correct and that the water hoses are not kinked, then cycle the keyswitch to clear the fault.

VSWR—turns on when either or both of the QSW RF connectors on the back of the power supply are disconnected, or when a connection fault causes poor coupling of RF power into the AOMs.

The **DIODE** LEDs turn on when a fault condition is detected in the diode laser pump chamber inside the laser head. These faults indicate a serious condition that is capable of causing permanent damage to the *Empower* if not quickly corrected.

OVERTEMP—turns on when the pump chamber exceeds the safe operating temperature. Turn off the power supply and allow the chiller to run for 10 minutes to cool it down. Then verify the chiller is set to the

correct temperature (18°C) and restart the laser. However, this time, run it at a lower diode laser drive current (which is set on the Setup menu).

OVERPOWER—turns on when diode pump laser power exceeds the factory-set limit. Reduce the diode laser drive current, then cycle the power on the power supply to clear the fault.

OVERVOLT—turns on when the voltage required to supply the requested current to the diode lasers exceeds the factory-set limit. Check the pump chamber umbilical connection, then cycle the ac power on the power supply to clear the fault.

OVERCURR—turns on when the current requested to drive the diode array exceeds the factory-set limit. Lower the current drive setting, then cycle power on the power supply to clear the fault.

The **DRIVER** LEDs turn on when a fault condition is detected in the diode lasers drive electronics in the *Model 2800*. These faults indicate a serious condition that is capable of causing permanent damage to the *Empower* if not quickly corrected.

PROTECTION—turns on when the internal drive FET protection circuit is activated.

Warning! 

Do not operate the *Empower* if the PROTECTION fault occurs! Contact your Spectra-Physics service representative immediately.

OVERTEMP—turns on when the diode laser driver (FET) heatsink becomes overheated due to inadequate airflow, or when a shorted or over-current condition is detected. Verify airflow through the power supply is adequate and allow the system to run without lasing for at least 10 minutes to cool it down, then restart the laser.

OVERPOWER—turns on when the internal drive FET exceeds a factory-set power limit. Reduce the diode drive current, then cycle the power on the power supply to clear the fault.

LBO TEMP LEDS turn on when communications between the computer and the power supply are interrupted.

TEMP—turns on when the LBO temperature is outside the factory set point. When this happens, the diode pump lasers turn off and, once off, cannot be turned on until the fault clears. Allow the temperature to return to specified temperature range before turning on the system again.

The LBO TEMP fault is triggered if the temperature of the doubling crystal becomes greater than ± 1°C from the pre-set optimum value.

STABILITY—turns on when the LBO temperature drifts outside of a range calculated from a running average. Allow the temperature to stabilize within the specified range before turning on the laser again. The oven should achieve LBO temperature stability within 10 minutes.

The LBO STABILITY fault works as follows. The software measures the temperature of the doubling crystal once per second and keeps a history of readings of the most recent 3 minutes (i.e., 180 readings). After each temperature reading, the software calculates the minimum and maximum of those 180 temperature readings. If the difference between the maximum and the minimum exceeds 2°C, a fault is issued. The diode pump lasers are turned off and, once off, they cannot be turned on until the fault clears.

Note



The LBO STABILITY indicator will illuminate for 3 minutes when the software is first started or restarted.

Chapter 7 Maintenance and Troubleshooting

The *Empower* laser system has been designed to require minimal maintenance. The routine procedures described in this chapter are recommended to keep the system in good working condition for the life of the laser. For further information, contact your Spectra-Physics service representative.

Routine Maintenance

Cleaning the Output Window

The *Empower* cavity is sealed and should never be opened to clean the interior optics. However, depending on the cleanliness of the environment, it is possible that dirt on the output windows may affect the performance of the laser. If this appears to be the case, contact your Spectra-Physics representative. The output window is difficult to clean. *Do not remove the cover without prior approval to do so, or you may void your warranty.*

Cooling Water Maintenance

To prevent metal corrosion and growth of algae in the *Empower* closed-loop cooling water system, OptiShield corrosion inhibitor is recommended. This chemical treatment prevents galvanic corrosion from dissimilar metals, prevents oxidation of ferrous metals and acts as an effective algaecide.

The first application of OptiShield is normally performed at the factory, and a similar application will be repeated during system installation. Thereafter, for most *Empower* systems the OptiShield solution should be changed once a year according to the following procedure.

Danger!



Before handling the OptiShield water treatment chemical, read the Material Safety Data Sheet included in Appendix A. This sheet describes the potential hazards of this material and provides handling precautions associated with it. *This chemical can be harmful if swallowed, inhaled or absorbed through the skin or eyes*.



Caution!



Use only **STEAM-DISTILLED** water in the *Empower* cooling system. The use of highly de-ionized water may cause corrosion damage.

Annual Changing of Solution

1. Shut down the *Empower* system and turn off the chiller.
2. Drain the old fluid from the chiller following the procedure in the chiller user's manual.
3. The chiller has a ½ gallon capacity. Refill the reservoir with ½ pint of OptiShield and the remaining volume with **steam-distilled** water.
4. Turn on the chiller and verify that water is flowing (e.g., the flow interlock has not been activated). Inspect for leaks at the hose connections between the laser head and chiller.
5. Verify the following chiller operating parameters:
   - a flow rate of at least 2.5 gal (9.5 l) per minute
   - water pressure between 50 and 55 psi (345 to 380 kPa)
   - water temperature between 65–86°F (18–30°C)

The chiller is now ready to resume operation.

Troubleshooting Guide

This general-purpose troubleshooting guide is provided to assist you in isolating some of the problems that might arise during use of the *Empower* laser system; for example, if performance drops unexpectedly. A complete repair procedure is beyond the scope of this manual and should be addressed by your authorized Spectra-Physics service representative.

Symptom: Laser will not start

Possible Cause:	Corrective Action:
There is no ac line power	Verify the power supply AC rocker switch is on. If it is, (a) verify the breaker on front of power supply is in the correct position, (b) verify the power cord connector at the rear of supply is tight, and (c) verify the fuse is good.
One or more interlocks are not closed	Check the FAULT LED on the power supply and the FAULT panel of the software. If these indicators are on, cycle the keyswitch off and on to clear latched interlocks. Investigate any interlocks that are not cleared by this action. For example, verify the chiller is turned on and that the hoses are not constricted; verify the flow direction is not reversed; verify the user interlock is in place; etc. Refer to Chapter 2 for more information on interlocks.
Keyswitch is not in the ON position	Turn the keyswitch to the ON position (horizontal). The key cannot be removed when it is in this position.
There is a communications error with the control computer	Verify there is a good connection with the serial cable. Exit the Empower control software, then turn off ac power to the power supply. Wait 5 seconds, then turn on power again and restart the control software.

Symptom: Emission light comes on, but there is no output

Possible Cause:	Corrective Action:
The output shutter is closed	The shutter operates automatically; recycle system power to clear a possible fault.
QSW MODE set to HOLD OFF	Change the setting of the QSW SOURCE selector in the SETUP screen of the Empower Control Software.
QSW SOURCE is set to EXTERNAL and no external signal is supplied.	Change the setting of QSW SOURCE to INTERNAL, or else supply an appropriate external trigger signal for the Q-switch.

Possible Cause:	Corrective Action:
The external Q-switch input is not valid	If an external Q-switch trigger is being used, verify the signal is at least 2 V into a 50-ohm load and the pulse is at least 2.5 µs wide. Also verify the frequency is the desired value and ±20% of the factory preset value and between 500 Hz and 10 kHz.
QSW RF driver cable(s) have been disconnected	If either Q-switch RF cable has been removed, the *Empower* will report a VSWR error and an interlock condition will occur. Verify both RF cables are attached at both the power supply and laser head.
Optics are out of alignment	Contact your authorized service representative.
The LBO crystal is damaged	Contact your authorized service representative.

Symptom: Output power varies

Possible Cause:	Corrective Action:
LBO temperature has drifted	Contact your authorized service representative.
Water temperature variation	Verify the water chiller temperature is set to the same setting used when it was installed (typically in the range of $18–30^{\circ}$C).
Algae growth is present in the cooling water	Perform the procedure in this chapter to replace the coolant.
Laser optics out of alignment	Contact your authorized service representative.
The output window is dirty	Perform the procedure in this chapter to clean the window.
Externally triggered Q-switch frequency is set higher or lower than the nominal value range	If operating the *Empower* at a repetition rate outside the nominal range, contact Spectra-Physics for the power specification at that frequency.
There is Q-switch breakthrough	With the current at the High setting, turn the QSW SOURCE selector to the HOLD OFF position and turn the laser on. Using an infrared viewer or card, verify there is no infrared and/or visible output beam from the laser. If infrared light is present, the Q-switch is malfunctioning or misaligned. Contact your authorized service representative.
Diode laser power is low	After several thousand hours of operation, the optical power of the diode pump lasers decreases and the current to the diode lasers must be increased to compensate. Contact your service representative.
The LBO crystal is damaged	Contact your authorized service representative.

Customer Service

At Spectra-Physics we take pride in the durability of our products. We place considerable emphasis on controlled manufacturing methods and quality control. Nevertheless, even the finest precision instruments need occasional service.

Warranty

This warranty supplements the warranty contained in the specific sales order. In the event of a conflict between documents, the terms and conditions of the sales order shall prevail.

The *Empower* is protected by a twelve-month warranty. All mechanical and optical parts and assemblies are unconditionally warranted to be free of defects in workmanship and material for the warranty period. At its election, Spectra-Physics will repair or replace without charge components that prove defective during the warranty period. The obligation of Spectra-Physics is limited to repair covered under the warranty return procedure described below. Equipment repaired or replaced is warranted only for the remaining original warranty period.

This warranty is in lieu of all other warranties, implied or expressed, and does not cover incidental or consequential loss.

Spectra-Physics will provide, at its expense, all parts and labor and one-way return shipping of the defective part or instrument (if required).

This warranty does not apply to any instrument or component not manufactured by Spectra-Physics. When products manufactured by others are included in Spectra-Physics equipment, the original manufacturer's warranty is extended to Spectra-Physics customers. When products manufactured by others are used in conjunction with Spectra-Physics equipment, this warranty is extended only to the equipment manufactured by Spectra-Physics.

This warranty does not apply to equipment or components that, upon inspection by Spectra-Physics, discloses to be defective or unworkable because of abuse, mishandling, misuse, alteration, negligence, improper installation, damage in transit, or other causes beyond the control of Spectra-Physics.

Return of the Instrument for Repair

Contact your nearest Spectra-Physics field sales office, service center or local distributor for shipping instructions or for an on-site service appointment. You are responsible for one-way shipment of the defective part or instrument to Spectra-Physics.

We encourage you to use the original shipping boxes. If the original boxes have been destroyed or lost, we recommend that you order new ones. We will return instruments only in Spectra-Physics containers.

Service Centers

Belgium

Telephone: (32) 0800 1 12 57

France

Telephone: (33) 0810 00 76 15

Germany and Export Countries*

Spectra-Physics GmbH
Guerickeweg 7
D-64291 Darmstadt
Telephone: (49) 06151 708-0
Fax: (49) 06151 79102

Japan (East)

Spectra-Physics KK
East Regional Office
Daiwa-Nakameguro Building
4-6-1 Nakameguro
Meguro-ku, Tokyo 153-0061
Telephone: (81) 3-3794-5511
Fax: (81) 3-3794-5510

Japan (West)

Spectra-Physics KK
West Regional Office
Nishi-honmachi Solar Building
3-1-43 Nishi-honmachi
Nishi-ku, Osaka 550-0005
Telephone: (81) 6-4390-6770
Fax: (81) 6-4390-2760

The Netherlands

Telephone: (31) 0900 5 55 56 78

United Kingdom

Telephone: (44) 1442-258100

United States and Export Countries**

Spectra-Physics
1330 Terra Bella Avenue
Mountain View, CA 94043
Telephone: (800) 456-2552 (Service) or
(800) SPL-LASER (Sales) or
(800) 775-5273 (Sales) or
(650) 961-2550 (Operator)
Fax: (650) 964-3584
e-mail: service@spectra-physics.com
sales@spectra-physics.com
Internet: www.spectra-physics.com

**And all European and Middle Eastern countries not included on this list.*
***And all non-European or Middle Eastern countries not included on this list.*

Appendix A Optishield Algaecide

This section contains the material safety data sheet supplied by the vendor for the algaecide used in the *Empower* closed loop cooling system. Please read this information before handling or disposing of the chemical or any cooling water containing it.

The information does not pertain to, and may not be appropriate for, larger quantities of the chemical than those described. The product is intended for use by personnel with sufficient technical skill and qualification to use the material correctly and understand the risks in handling any potentially hazardous chemical.

Warning! 

This information, obtained from the manufacturer, is believed to be reliable and accurate but has not been verified independently by Spectra-Physics. Accordingly, no representation or warranty, expressed or implied, with respect to merchantability and fitness for a particular purpose is made in this data sheet. For further information, contact the manufacturer at the address listed.

 **Caution!** 

Use only **STEAM-DISTILLED** water in the *Empower* cooling system. The use of highly de-ionized water may cause corrosion damage.

Filling the Chiller for First Time Use

Refer to the chiller user manual for information on how to fill the reservoir.

1. Flush the cooling circuit with distilled water.
2. The chiller reservoir has a ½ gallon capacity. Fill it with 4 ounces of Optishield® fluid and the remaining volume with **steam-distilled** water.
3. Circulate this solution for about 30 minutes.
4. Drain the fluid. **Do not rinse.** Continue with Step 2 under "Periodic Changing of Solution" on page A-2.

OptiShield is a registered trademark of Opti Temp, Inc.

Periodic Changing of Solution

Refer to the chiller user manual for information on how to drain the unit.

1. Drain the coolant.
2. Refill the reservoir with ½ pint of OptiShield fluid and the remaining volume with **steam-distilled** water.
3. Turn on the chiller and verify that water is flowing. Inspect for leaks at the hose connections between the laser head and chiller.
4. Turn off the chiller.

Please note that it takes the chiller about 15 minutes to stabilize the temperature of the laser head cold plate and, thus, the output of the laser. If the laser is used frequently, leave the chiller on between periods of use to eliminate this stabilization period. If the laser is used infrequently, turn the chiller off between periods of use.

MSDS Sheet for Optishield

MATERIAL SAFETY DATA SHEET

OPTISHIELD®

DATE REVIEWED: July, 2001

DISTRIBUTOR:
OPTI TEMP INC.
P.O. Box 5246
Traverse City, Michigan
49696
(Collect)

Telephone: 231-946-2931
Emergency Hot Line
Chem-Tel 800-255-3924
International 813-977-3668

USE OF PRODUCT: Treatment of select cooling water systems.

SECTION I - Product Identification

Special Components Chemical Names
This is a proprietary blend which contains the following components:

Name	% In Product
Sodium Nitrate	<1%
Sodium Molybdate	1-2%
Sodium Hydroxide	1-2%
Phosphoric Acid	<1.5%
Triazole	<2.5%

The Hazardous Materials Index Rating is as follows: Health=1; Flamability=0; Reactivity=1

SECTION II- Physical Data
% Volatiles: Nil
Color: Light Golden-Yellow
Odor: Mild
Specific Gravity: 1.03
Solubility: 100% in water
pH of 100% solution (concentrated) 11 to 12; typical 11.5

SECTION III - Fire and Explosion Hazard
Non-Flammable
Flash Point: Unknown

SECTION IV - Reactivity Data
Stability: Stable
Polymerization: None
Exposure to Other Chemicals: Keep away from concentrated acids
Reactivity in Water: None

SECTION V - Shipping Information
Regulation: This material is not a DOT regulated material.
This product is a freezable liquid when and where applicable.

SECTION VI –Spill
Small spills: Small spills may be soaked up using common absorbent material, and using appropriate safety equipment. Dispose of and handle in accordance with local, state, and federal regulations.

Large spills: Large spills should be pumped into suitable containers located in diked areas. Residual material should be cleaned up with water. Dispose of and handle in accordance with local, state, and federal regulations.

SECTION VII - First Aid
Ingestion: Give milk or water, induce vomiting, get medical attention.

Skin: Flush with fresh water, wash with soap and water. Remove contaminated clothes and shoes.

Eyes: Flush with fresh water for at least 15 minutes. Get medical attention.

Inhalation: Inhalation should not occur during normal operation. However, should it occur, close container and move to well-ventilated area. If irritation persists, get medical attention.

SECTION VIII - Special Instructions
Do **NOT** pressurize container.

Keep container closed at all times when not in use.

Store in cool area above 60˚F. Do not allow fluid to freeze.

Use in well-ventilated area. Do not breathe mist or vapor.

Wash hands thoroughly after handling product.

Protect eyes with safety goggles or glasses with side shields.

Notes

Report Form for Problems and Solutions

We have provided this form to encourage you to tell us about any difficulties you have experienced in using your Spectra-Physics instrument or its manual—problems that did not require a formal call or letter to our service department, but that you feel should be remedied. We are always interested in improving our products and manuals, and we appreciate all suggestions.

Thank you.

From:

Name _____

Company or Institution _____

Department _____

Address _____

_____

Instrument Model Number _____ Serial Number _____

Problem: _____

_____

_____

_____

_____

Suggested Solution(s): _____

_____

_____

Mail To:

Spectra-Physics, Inc.
Quality Manager
1335 Terra Bella Avenue, M/S 15-50
Post Office Box 7013
Mountain View, CA 94039-7013
U.S.A.

E-mail: sales@spectra-physics.com
www.spectra-physics.com

FAX to:

Attention: Quality Manager
(650) 961-7101